

What Goes Up Must Not Come down - Time Series Momentum in Factor Risk Premiums

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Abstract

I document significant time variation and predictability in a set of risk factors based on technical indicators. As these indicators are primarily designed to detect trends in asset prices, these findings imply substantial time series momentum in factor risk premiums. Specifically, risk premiums are significantly larger (lower) following recent uptrends (downtrends) in the underlying risk factor. A trend-based dynamic factor strategy, which uses the trend-based signals in order to lever up or down the risk factor, yields annual utility gains of up to 500 basis points, doubles the risk factors' Sharpe ratio, and more than halves their maximum drawdown. In a conditional asset pricing context, employing technical indicators as conditioning information substantially improves a model's explanatory power. Overall, my evidence poses several challenges to risk-based asset pricing theories and seems to be more in line with theories of sentiment and investors initial under- and subsequent overreaction to new information.

JEL classification: G11, G12, G14, G17.

Keywords: time-varying risk premiums, return predictability, time series momentum, dynamic risk factor strategy, conditional asset pricing, sentiment.

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1. Introduction

In the center of asset pricing lie various risk factors as driving forces of an asset's expected return. The latest five-factor model of Fama and French (2014) (FF5), for example, states that the market factor (RMRF), and several risk factors related to "salient" firm characteristics, such as size (small minus big market capitalization, SMB), value (high minus low book-to-market ratio, HML), profitability (robust minus weak profitability, RMW) and investment (conservative minus aggressive investments, CMA) explain the cross-section of expected stock returns. However, with an exception of the equity risk premium, little is known about the time variation in factor risk premiums and to what extent this potential time variation is predictable using publicly known information. As emphasized in Cochrane (2011), answers to these questions are of great interest to academics and practitioners alike. On the academic side, understanding the time variation in factor risk premiums could offer new insights into market efficiency and enhance existing asset pricing models. On the practitioners' side, investment as well as financing decisions could be improved.

In two recent studies, Moskowitz et al. (2012) and Neely et al. (2014) document a statistically and economically significant predictive ability of technical indicators for the equity risk premium, which even exceeds that of macroeconomic variables. By construction, technical indicators are designed to identify trends in asset prices based on past price information. These price trends are then believed to persist in the future, resulting in time series momentum in the underlying asset.¹ Hence, insofar as asset prices are not pure random walks but exhibit periodic trends, i.e. time series momentum,² technical indicators should prove informative.

In this study, I examine the time variation and predictability based on technical indicators in factor risk premiums associated with the FF5 factors as well as the momentum factor (WML) of Jegadeesh and Titman (1993). Using data spanning from July 1963 to December 2015, I find a strongly statistically and economically significant predictive ability of technical indicators for aforementioned risk factors, both in- and out-of-sample. Hence, the time series momentum effect documented in Moskowitz et al. (2012) and Neely et al. (2014) is not only prevalent in the equity risk premium, but also applies to other factor risk premiums. More specifically, risk premiums are substantially larger (lower) following recent uptrends (downtrends) in the underlying risk factor with a statistically significant return difference between the two states of up to 14.03% per annum, implying an economically sub-

¹Time series momentum is related to, but different from, the phenomenon known as "momentum" in the finance literature. Momentum is cross-sectional in nature as it focuses on the relative performance of assets in the cross-section, whereas time series momentum focuses purely on an asset's own past return.

²In the remainder of the text, I use the term "trend" interchangeably with "time series momentum".

stantial time variation. Moreover, the predictability is strongest during factor downtrends and crashes when it is most useful to the representative investor. These results are robust to different parameterizations of the considered technical indicators, using various forecasting combination methods, apply to both legs of the risk factors, are stable over time and are not a result of data mining.

Using these insights, I introduce optimal trend-based dynamic factor strategies, in which the risk factor is levered up or down over time based on a technical indicator's trend-based signals, so as to maximize its unconditional Sharpe ratio. Intuitively, an investor should hold an asset only when its price is on an uninterrupted uptrend, which may be due to a host of factors known or unknown to the investor. However, when the trend is broken, the investor should sell the asset as new factors may come into play. This approach is inspired by the dynamic momentum strategy of Daniel and Moskowitz (2016). However, in contrast to their strategy, mine is applicable on all risk factors and is easily implementable in real time. In the empirical analysis, I document that trend-based dynamic factor strategies significantly outperform the standard static factor strategies and those using forecasts based on their historical average return. More specifically, they yield sizeable utility gains for mean-variance investors of up to 500 basis points, more than double the risk factors' Sharpe ratio and halve their maximum drawdowns. Contradicting conventional risk-based explanations, these substantial improvements appear not to be a compensation for crash risk or tail risk. In contrast, trend-based dynamic strategies rather provide insurance for extreme events, as their relative returns tend to be largest when risk factor returns are most extreme - with its best relative performance during periods of crashes. For example, during the prominent "momentum crash" of 2009, which constitutes the biggest drawdown in any risk factor observed over my sample period, the trend-based dynamic strategy applied on WML reduced its losses from -52.3% to -15.6% between October 2008 and October 2009. Technical indicators prove informative during extreme events, because crashes often happen when the risk factor goes from normal to bad (resulting in the start of a downtrend, a sell signal of the technical indicator and a low weight in the risk factor within the dynamic strategy), and then from bad to worse (leading the trend-based dynamic strategy to prevent large losses).

The documented strong and predictable time variation in factor risk premiums suggests linking these findings to the cross-section of expected stock returns. It is well known that allowing for time variation in factor loadings can only improve the explanatory power of asset pricing models, if the loadings also covary with the corresponding factor risk premium or factor volatility. Hence, I use trend-based signals issued by the technical indicator as conditioning variable and introduce trend-based conditional versions of the most popular asset pricing models: the Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner

(1965), the three-factor model of Fama and French (1993) (FF3) and the five-factor model of Fama and French (2014) (FF5).

Using test assets that pose particular challenges for existing asset pricing models, such as size and momentum (Fama and French (2015)) as well as industry-sorted portfolios (Fama and French (1997)), I document significant improvements in explanatory power for trend-based conditional asset pricing models. Specifically, the trend-based conditional CAPM explains about 60% of the cross-sectional variation in size and momentum sorted returns compared to 23% of the unconditional CAPM, matching the explanatory power of the unconditional FF3 and FF5. A further significant improvement in explanatory power to about 84% and 83% is observed for the trend-based conditional FF3 and FF5, respectively. And even more remarkable, the zero-beta rate in the cross-sectional asset pricing test becomes statistically insignificant for both trend-based conditional models. With respect to industry-sorted portfolios, the improvements in explanatory power are even more pronounced. The success of the trend-based conditional models is primarily due to the fact that test assets with positive unconditional pricing errors, like winner portfolios for example, exhibit higher (lower) conditional factor loadings particularly following uptrends (downtrends) in the corresponding risk factors, when risk premiums are particularly high (low), and vice versa for test assets with negative unconditional pricing errors. These results provide additional evidence for the genuine economic information provided by technical indicators and highlight the importance of accounting for time series momentum in factor risk premiums when determining an asset's expected return.

My finding of time series momentum in a set of factor risk premiums contradicts the "random walk" hypothesis, which in its basic form implies that knowing an asset's past return should not be informative about its future expected return. While a rejection of the random walk hypothesis does not necessarily imply a rejection of a more sophisticated notion of market efficiency with time-varying risk premiums reflecting the time-varying compensation for bearing aggregate risk, my empirical results also pose several challenges to risk-based asset pricing theories. First, if one makes the straightforward assumption that recent prolonged uptrends (downtrends) in a risk factor are due to positive (negative) news and hence represent improving (deteriorating) economic times with decreasing (increasing) marginal utility, the finding of significant higher (lower) risk premiums following recent uptrends (downtrends) contradicts the risk-based explanation proposed for the time variation in factor risk premiums (see, for example, Fama (1991)). Specifically, the risk-based explanation argues that risk premiums should be high (low) during bad (good) economic times when aggregate risk aversion, aggregate risk, and consequently also marginal utility are high (low). Supporting the inconsistency of the documented predictability with time-varying compensation for ag-

gregate risk, I show low and diverse pairwise correlations between the factor risk premiums as implied by their respective trend-based signals as well as the trend-based dynamic factor strategies. Second, I find higher (lower) factor risk as measured by factor volatilities following uptrends (downtrends) in the respective risk factors when risk premiums are high (low), which contradicts conventional risk-based asset pricing theories suggesting that the traditional risk-return trade-off should be positive as in Merton (1980) and French et al. (1987). And third, the fact that the trend-based dynamic factor strategies significantly improve the risk factors' Sharpe ratio, substantially reduce their drawdowns and provide insurance for extreme events, questions the risk-based explanation proposed for the existence of factor risk premiums, which relies on occasional large losses particularly during bad economic times, when marginal utility is high (see, for example, Fama and French (1993)).

Now, what explains these results? In the empirical analysis, I provide evidence that rational theories of momentum based on a serial correlation in time-varying expected returns due to risky growth options (Berk et al. (1999), Johnson (2002), and Sagi and Seasholes (2007)) seems unlikely to explain my results. Likewise, I don't find any relation between the trend-based signals or the returns on trend-based dynamic factor strategies and aggregate liquidity risk. In contrast, my findings seem to be more consistent with theories of sentiment and an informational inefficient market in which investors' initial under- and subsequent overreaction to new information leads to the observed time series momentum in factor risk premiums (Barberis et al. (1998), (Daniel et al., 1998), and Hong and Stein (1999), among many others).³ This effect can eventually be further intensified by informational uncertainty (Zhang (2006)), investor sentiment (Baker and Wurgler (2006, 2007)), feedback traders (De Long et al. (1990)), feedback effects that actually alter firm fundamentals (Edmans et al. (2015)) or a gradual market response to new information due to investors receiving information at different points in time or processing information at different speeds (Chan et al. (1996)). Due to limits to arbitrage, the caused mispricing then might persist, even if rational investors recognize the deviations from fundamental value (Shleifer and Vishny (1997)). In line with this behavioral explanation, I find stronger time series momentum in the leg of the risk factor that covers stocks that are more difficult to value and hence, more sensitive to speculative demand, more affected by investor sentiment and informational uncertainty, as well as more costly to arbitrage. In addition, I provide evidence that the documented time series momentum in factor risk premiums is partly driven by the effects of investor sentiment on the cross-section of stock returns.

My study relates to the vast literature on the time variation and predictability of factor

³Initial under- and subsequent overreaction can be due to a plethora of cognitive and behavioral biases (see Footnote 2 in Moskowitz et al. (2012)).

risk premiums. This literature generally documents the predictive ability of several macroeconomic or technical variables for the equity risk premium, with little evidence for other factor risk premiums despite their increasing importance in the asset pricing literature (see Rapach and Zhou (2013) for a recent survey). And if predictability is documented in factor risk premiums other than the equity risk premium, it is restricted to one specific predictor variable for one specific risk factor.⁴ I complement this literature by showing that technical indicators exhibit strong predictive ability for a set of risk factors, both in- and out-of-sample. The fact that one predictor variable can be used to predict the time variation in a set of factor risk premiums, thereby, guards against the potential issue of data mining. In addition, by investigating the economic sources driving the documented predictability, this paper adds to the debate on whether risk or behavioral biases explain the patterns in asset price behavior.

By introducing a trend-based dynamic factor strategy, my study also contributes to the literature on dynamic risk factor strategies (Grundy and Martin (2001), Barroso and Santa-Clara (2015), and Daniel and Moskowitz (2016)). However, in contrast to the aforementioned studies, which focus exclusively on the momentum factor, my trend-based dynamic factor strategy is applicable on all risk factors and easily implementable in real time.

Finally, my paper is also related to the literature on conditional asset pricing models starting with Jagannathan and Wang (1996) and Lettau and Ludvigson (2001). However, existing conditional models typically employ persistent macroeconomic variables that are only available at monthly or quarterly frequency and, due to the lack of evidence on the predictability of factor risk premiums other than the equity risk premium, only introduce conditional versions of the simple or consumption CAPM. I complement this literature by introducing trend-based conditional asset pricing models that allow for time variation in factor loadings and factor risk premiums at higher frequencies and are also applicable to other asset pricing models such as the FF3 or FF5. In addition, my empirical results add to the literature that tries to explain the high abnormal returns on momentum- (see, for example, Grundy and Martin (2001) and Fama and French (2015)) as well as industry-sorted portfolios (Fama and French (1997)) in the context of an asset pricing model.

The rest of the paper is organized as follows. Section 2 describes our data set, including the construction of the technical indicators. Section 3 reports in-sample and out-of-sample

⁴With respect to the size premium, Hou (2007) find evidence on the predictability of small firm returns based on lagged large firm returns, while Swaminathan (1996), Neal and Wheatley (1998), and Lemmon and Portniaguina (2006) present evidence on the predictability of the size premium based on different measures of investor sentiment. With respect to the value premium, Lewellen and Nagel (2006) find that it is not predictable, while Stivers and Sun (2010) document positive evidence on its predictability based on the market's recent cross-sectional dispersion in stock returns.

predictive regression results, while Section 4 reports results for the trend-based dynamic factor strategies. Section 5 provides results for the conditional asset pricing tests, and Section 6 investigates potential explanations for the documented predictability. Section 7 concludes.

2. Technical Indicators

In this section, I introduce the risk factors and technical indicators considered in this study and present some summary statistics.

2.1. Risk Factors

The choice of risk factors is motivated by the intention to cover a parsimonious set of factors that are able to explain the cross-section of expected stock returns. In two recent studies, Fama and French (2008, 2015) demonstrate the strong explanatory power of their FF3 and their extended FF5 for the majority of return anomalies documented in the prior literature. Hence, I consider the factors of the FF5: RMRF, SMB, HML, RMW, and CMA. Because the momentum anomaly introduced by Jegadeesh and Titman (1993) remains a pervasive anomaly in the context of the aforementioned asset pricing models (Fama and French (2008, 2015)), I also add WML to the set of considered risk factors. Daily returns on these risk factors are obtained from Kenneth French's data library. The sample period ranges from July 1963 to December 2015.

2.2. Construction of Technical Indicators

Following the recent literature (see, for example, Neely et al. (2014), Moskowitz et al. (2012), and Han et al. (2013)), I focus my analysis on two trend-following strategies that are representative of the wide range of technical indicators and are also widely used by market participants: moving average (MA) and time series momentum (MOM) rules.

A MA rule generates a buy or sell signal ($S_{i,t} = 1$ or $S_{i,t} = 0$, respectively) at the end of period t for asset i by comparing two moving averages of different lag length:

$$S_{i,t} = \begin{cases} 1 & \text{if } MA_{i,t}^s \geq MA_{i,t}^l, \\ 0 & \text{if } MA_{i,t}^s < MA_{i,t}^l, \end{cases} \quad (1)$$

where

$$\text{MA}_{i,t}^j = (1/j) \sum_{u=0}^{j-1} P_{i,t-u}, \quad (2)$$

where $P_{i,t}$ is the price of asset i at the end of period t , and s (l) is the lag length of the short (long) MA ($s < l$), i.e. the number of past periods, including period t , considered in the calculation of the short (long) MA. In the remainder of this study, I denote the MA rule with lag lengths s and l on asset i by $\text{MA}_i(s, l)$. The MA rule is able to detect recent price trends because, by construction, the short MA will be more sensitive to recent price changes and as a result will be above (below) its longer counterpart following a recent uptrend (downtrend) in the price of the asset. Hence, buy (sell) signals generated by the MA rule indicate recent uptrends (downtrends) in the price of the underlying asset. In order to account for potential data mining concerns, I examine seven MA strategies with different parameterization: $\text{MA}(1, 20)$, $\text{MA}(1, 40)$, $\text{MA}(1, 120)$, $\text{MA}(1, 180)$, $\text{MA}(1, 240)$, $\text{MA}(20, 180)$, and $\text{MA}(20, 240)$. Intuitively, a change in the lag length modifies a strategy's sensitivity to recent price changes. More specifically, a shorter (longer) lag length, on the one hand, allows the MA rule to react faster (slower) to trend reversals and thereby generally increases (decreases) its profitability. On the other hand, however, it also increases (decreases) the frequency of changes in the signal, i.e. the amount of trading required to implement the signals, and in turn also potential transaction costs.

A time series momentum rule generates signals by comparing the current price of asset i to its price m periods ago:

$$S_{i,t} = \begin{cases} 1 & \text{if } P_{i,t} \geq P_{i,t-m}, \\ 0 & \text{if } P_{i,t} < P_{i,t-m}. \end{cases} \quad (3)$$

Intuitively, a higher (lower) current price indicates positive (negative) momentum and, therefore, generates a buy (sell) signal. In the remainder of this study, I denote the time series momentum rule with lag length m on asset i by $\text{MOM}_i(m)$, and compute signals for $m = 180, 240$.

The idea behind both technical indicators is that an investor should only hold an asset when the asset price is on an uninterrupted uptrend ($S_{i,t} = 1$), sell the asset if the trend is broken ($S_{i,t}$ changes from 1 to 0) and stay out of the asset as long as the asset is on an uninterrupted downtrend ($S_{i,t} = 0$). Hence, insofar as risk factors are not pure random walks but exhibit period trends, i.e. time series momentum, technical indicators should prove informative, because they are primarily designed to detect trends in asset prices.

In sum, this leaves me with nine individual technical indicators. However, in order

to simplify the empirical analysis, I focus on the MA(1,180) in the main section of this study and examine the sensitivity of my main results to the other individual indicators, principal components extracted from the entire set of individual indicators and different combination methods for individual forecasts proposed by Rapach et al. (2010) in Section C of the Appendix. In the remainder of this study, I denote the MA(1,180) rule by TI. The choice of TI's lag length is motivated by the prior literature. Moskowitz et al. (2012) show that time series momentum in equity index, currency, commodity, and bond futures is strongest for look-back periods between 9 to 12 months, whereas shorter horizons of up to one month (Jegadeesh (1990)) and longer horizons of more than one year (De Bondt and Thaler (1985)) are generally characterized by return reversals.

2.3. *Summary Statistics*

Panel A of Table 1 provides summary statistics on the set of considered risk factors such as their annualized mean return, annualized standard deviation, Sharpe ratio, minimum as well as maximum one-day return, and daily autocorrelation. The statistics are calculated over the entire sample period from April 1964 through December 2015. Overall the results are consistent with recent empirical evidence.

[TABLE 1 ABOUT HERE]

Summary statistics on the trend-based signals of TI applied on the different risk factors are reported in Panel B. The strategies produce buy signals ($S_{i,t} = 1$) between 52% and 72% of all trading days and the average holding period for each position, i.e. the average number of days until the signal changes, varies between 60 and 100 days. As a result, the first order autocorrelation of the trend-based signals is close to one. These results already give a first indication of the amount of trading required to implement the trend-based signals.

3. Predictive Regression Analysis

In this section, I study the time variation in factors risk premiums by investigating TI's ability to predict the returns on the aforementioned set of risk factors, both in- and out-of-sample. In addition, I examine how this potential predictive ability evolved over time. These results then allow first conclusions about the existence of time series momentum in factor risk premiums as technical indicators are primarily constructed to detect trends in asset prices.

3.1. In-Sample Analysis

The standard framework for examining return predictability is the following predictive regression model:

$$r_{i,t+1} = \alpha_i + \beta_i S_{i,t} + \epsilon_{i,t+1}, \quad (4)$$

where $r_{i,t+1}$ is the return on asset i during period $t + 1$ and $S_{i,t}$ is the trend-based signal applied on asset i available at the end of period t . In this framework, the null hypothesis of interest is that $S_{i,t}$ has no predictive ability, $\beta_i = 0$. In this case, Equation (4) reduces to the constant expected return model, $r_{i,t+1} = \alpha_i + \epsilon_{i,t+1}$. In order to increase the power of the test, Inoue and Kilian (2004) recommend using a one-sided alternative hypothesis, as theory often suggests the sign of β_i under predictability. Hence, because $S_{i,t} = 1$ ($S_{i,t} = 0$) represents a buy (sell) signal, I test $H_0 : \beta_i = 0$ against $H_A : \beta_i > 0$ using ordinary least squares (OLS) with heteroskedasticity-consistent t -statistics.

Statistical inferences in Equation (4) are complicated by at least two reasons. First, as technical indicators, like most predictor variables, are highly persistent, there is a potential concern about a spurious regression bias as discussed in Ferson et al. (2003). And second, if the predictor variable is highly persistent and also correlated with the dependent variable, the well-known Stambaugh (1999) small-sample bias potentially distorts the coefficient estimates in a finite sample and inflates the associated t -statistics.

In order to alleviate these potential concerns, I employ two strategies to test $H_0 : \beta_i = 0$ against $H_A : \beta_i > 0$. First, I follow Neely et al. (2014) calculate empirical p -values using a wild bootstrap procedure that accounts for the persistence in predictor variables, correlations between the dependent variable and predictor innovations as well as general forms of return distribution. Second, I also report alternative asymptotic statistical inferences based on a powerful Wald test introduced by Kostakis et al. (2015) that is robust to the degree of persistence in the predictor variables (unit root, local-to-unit root, near stationary, or stationary).⁵

Overall, the results shown in Table 2 reveal a strong and positive predictive ability of TI over the entire sample period for all considered risk factors. While the regression intercepts, α_i , are generally statistically insignificant, the regression slopes, β_i , are strongly statistically significant at the 1% level (except for RMRF which is significant at the 5% level) based on both bootstrapped p -values and the IVX-Wald statistic of Kostakis et al. (2015). Hence, these results imply significant time series momentum in factor risk premiums. Specifically, risk premiums are significantly larger (lower) following recent uptrends (downtrends) in the underlying risk factor.

⁵See the cited articles for a detailed explanation of the two procedures.

Intuitively, the regression intercept, α_i , can be interpreted as the conditional expected return following downtrends ($S_{i,t} = 0$), while the regression slope, β_i , captures the expected return difference between up- and downtrends. Consequently, the conditional expected return following uptrends ($S_{i,t} = 1$) is equal to the sum of both components. Consider HML for example. Following downtrends, its conditional expected return is equal to -3.46% per annum, while it is 9.57% following uptrends. Comparing these estimates to its unconditional expected return of 3.94% documented in Table 1 implies a substantial pro-cyclical time variation in the value premium and a strong economic significance of TI's predictive ability. Similar results apply to all other considered risk factors. Also note that even factor risk premiums that seem to have vanished over time, like SMB (see van Dijk (2011) for a recent survey), may still be economically important in a conditional context.

[TABLE 2 ABOUT HERE]

At first glance, the R^2 statistics of Equation (4) reported in Table 2 appear small. However, because of the large unpredictable component in daily stock returns and the use of a dummy variable as the only explanatory variable in the predictive regression model, the R^2 will necessarily be very small. Following Campbell and Thompson (2008), even a daily R^2 of around 0.1% can translate into substantial utility gains for risk-averse investors. Except for RMRF and WML, all risk factors surpass this threshold. Moreover, the predictive ability of TI seems to be weakest for RMRF and strongest for HML.

From an economic point of view, it is also important to examine TI's predictive ability during different states to get a better understanding of its fundamental driving forces. Hence, following the literature (see Han et al. (2013) and Neely et al. (2014) among others), I calculate the R^2 statistics separately for two different states,

$$R_{i,c}^2 = 1 - \frac{\sum_{t=1}^T I_{i,t}^c \hat{\epsilon}_{i,t}^2}{\sum_{t=1}^T I_{i,t}^c (r_{i,t} - \bar{r}_i)^2}, \quad (5)$$

where $I_{i,t}^c$ is an indicator variable that is equal to one when asset i in period t is in state c and zero otherwise, $\hat{\epsilon}_{i,t}$ is the fitted residual based on the full-sample estimates of the predictive regression model in Equation (4), $\bar{r}_i$ is the full-sample mean of the return on asset i and T is the number of observations in the sample. According to Neely et al. (2014) the predictive ability of technical indicators on the equity risk premium is highly concentrated during National Bureau of Economic Research (NBER)-dated business cycle recessions. Columns six and seven in the first row of Table 2 confirm these results. However, for other risk factors, TI's predictive ability is generally stronger during NBER-dated business cycle expansions instead. Using recent price trends as an additional state indicator reveals why. Following Han et al.

(2013), columns eight and nine report R^2 statistics separately for up- and downtrends in the underlying risk factor defined based on whether the risk factor's return during the previous year was positive or not. The results reveal a stronger predictive ability during downtrends (except for SMB). Hence, with respect to TI's predictive ability it is less important whether the economy is in a recession or expansion, but instead more relevant whether the underlying risk factor is in an up- or downtrend. It is only for RMRF that NBER-dated business cycle recessions coincide with large negative returns, i.e. downtrends, in the respective risk factor. These results generalize the conclusion in Neely et al. (2014) to other risk factors and highlight the crash insurance characteristic of technical indicators which is particularly valuable from an asset allocation perspective as marginal utility is generally higher during downtrends in a given risk factor. I will investigate the crash insurance characteristic of technical indicators more thoroughly in the following sections of this paper.

In my analysis, I don't control for any other potential predictor variables. This choice is supported by at least five reasons. First, in an unreported tests, I find that the predictor variables considered in Goyal and Welch (2008), if at all, only exhibit predictive ability for the equity risk premium. Second, evidence on predictability in factor risk premiums other than the equity risk premium is weak (see Rapach and Zhou (2013) for a recent survey on return predictability). If at all, studies document the predictive ability of one specific predictor variable for one specific risk factor. Third, Neely et al. (2014) show that technical indicators and macroeconomic variables provide complementary information relevant for predicting the equity risk premium. Hence, in their study, accounting for macroeconomic variables does not affect the predictive ability of technical indicators. Fourth, Table A1 in the Appendix shows low and diverse pairwise correlations of the trend-based signals ranging between -0.23 (SMB and RMW) and 0.41 (HML and CMA). A universal macroeconomic variable is, therefore, unlikely to explain the documented predictability. Fifth, most other potential predictor variables are only available on monthly or even lower frequencies.

3.2. Out-of-Sample Analysis

Even though in-sample predictive regressions provide more efficient parameter estimates by considering the entire sample period, Goyal and Welch (2008) argue that a given return predictor will only be of economic significance, if it can sustain its level of predictability out-of-sample.

Hence, in order to run out-of-sample tests, I start by recursively calculating the period $t + 1$ out-of-sample trend-based return forecast for asset i based on the predictive regression in Equation (4) as

$$\hat{r}_{i,t+1}^{TI} = \hat{\alpha}_{i,t} + \hat{\beta}_{i,t} \text{TI}_{i,t} + \epsilon_{i,t+1}, \quad (6)$$

where $\hat{\alpha}_t$ and $\hat{\beta}_t$ are the OLS estimates of the regression coefficients based on data from the beginning of the sample in April 1964 through date t in order to avoid any potential look-ahead-bias. For the out-of-sample evaluation, I then divide the total sample period into an initial in-sample estimation period comprised of the first n_1 observations and an out-of-sample period comprised of the last $n_2 = T - n_1$ observations. I use the period from April 1964 to April 1969 as the initial estimation period, so that the out-of-sample evaluation period spans from May 1969 to December 2015. The length of the initial estimation period is motivated by the intention to balance having enough observations for reasonably precisely estimated initial forecasts with the desire for a relatively long out-of-sample evaluation period. As benchmark model, I employ the historical average forecast, which is based on the average return on risk factor i from the beginning of the sample through date t :

$$\hat{r}_{i,t+1}^{\text{HA}} = \frac{1}{t} \sum_{s=1}^t r_{i,s}. \quad (7)$$

The historical average forecast serves as a natural benchmark, as it corresponds to the constant expected return model in Equation (4) with $\beta = 0$, and implies that returns are unpredictable. Moreover, Goyal and Welch (2008) show that the historical average is a very stringent out-of-sample benchmark, as forecasts based on individual macroeconomic variables typically fail to outperform it when predicting the equity risk premium.

To compare the competing forecasts, I employ the widely used Campbell and Thompson (2008) out-of-sample R_{OOS}^2 statistic, the Clark and West (2007) *MSFE-adjusted* statistic, the Diebold and Mariano (1995) t -statistic modified by McCracken (2007), and the Clark and McCracken (2012) modified version of the White (2000) reality check. Intuitively, the R_{OOS}^2 statistic measures the proportional reduction in the mean squared forecast error (MSFE) for the predictive regression forecast relative to the historical average benchmark. A positive value of R_{OOS}^2 indicates a superior predictive ability of TI in terms of MSFE, whereas a negative value signals the opposite. While the R_{OOS}^2 measures the relative improvement in the MSFE, it is also important to investigate whether this potential improvement is statistically significant: that is, to test the null hypothesis that the historical average MSFE is less than or equal to the trend-based MSFE against the one-sided (upper-tail) alternative hypothesis (corresponding to $H_0 : R_{OOS}^2 \leq 0$ against $H_A : R_{OOS}^2 > 0$). In order to test the aforementioned null hypotheses, I follow the literature and utilize the *MSFE-adjusted* statistic of Clark and West (2007) (CW-test hereafter) and the Diebold and Mariano (1995) t -statistic modified by McCracken (2007) (DM-test hereafter).

All in all, the out-of-sample results reported in Table 3 confirm the previous in-sample results. In terms of MSFE, the trend-based forecasts outperform the historical average benchmark for each risk factor. The R^2_{OOS} statistics are all positive ranging between 0.01% and 0.25%. Based on both the CW- as well as the DM-test, these improvements in the MSFEs are also strongly statistically significant. Similar to the in-sample results, the predictive ability seems to be weakest for RMRF while it seems to be strongest for HML. Also matching the in-sample results, the out-of-sample predictive ability of TI is uniformly stronger following downtrends in the underlying risk factor.

[TABLE 3 ABOUT HERE]

The last four columns of Table 3 provide a sense of potential bias-efficiency trade-offs in the competing forecasts by reporting the Theil (1971) MSFE decomposition into the squared forecast bias and a remainder term.⁶ Overall, trend-based forecasts appear less biased and slightly more efficient than the historical average benchmark.

The prior literature has pointed out potential sources of data mining when examining stock return predictability (see, for example, Ferson et al. (2003)). This concern is particularly pronounced for technical indicators, as their dependence on a set of parameters leaves much room for potential data mining. Hence, in order to allay these concerns, I follow the literature (see, for example, Neely et al. (2014)) and employ the Clark and McCracken (2012) modified version of the White (2000) reality check, which is appropriate to compare a large set of forecasts from different models, all of which nest a benchmark model. More specifically, for each risk factor, we use the Clark and McCracken (2012) $maxMSFE-F$ statistic to test the null hypothesis that the historical average MSFE is less than or equal to the minimum MSFE of trend-based forecasts based on all the nine technical indicators introduced in Section 2.2. Again statistical inferences are based on wild fixed-regressor bootstraps. Except for RMRF, the null hypothesis is rejected at the 1% level, indicating that data mining cannot readily explain the significant out-of-sample predictive ability of TI. Detailed results for the alternative technical indicators introduced in Section 2.2, principal components extracted from the entire set of technical indicators and trend-based forecasts based on various combination methods introduced in Rapach et al. (2010) are reported in Section C of the Appendix.

⁶Following Theil (1971), the decomposition of the MSFE is given by $MSFE = (\hat{r}_i - \bar{r}_i)^2 + (\delta_{\hat{r}_i} - \rho\delta_{r_i})^2$, where $(\hat{r} - \bar{r})^2$ is the squared bias and the rest of the equation is the remainder term. More specifically, $\bar{r}_i$ ($\hat{r}_i$) is the mean of the actual (forecasted) daily return on asset i based on a given model, δ_{r_i} ($\delta_{\hat{r}_i}$), is the standard deviation of the actual (forecasted) daily return, and ρ is the correlation coefficient between the actual and the forecasted daily return, each computed over the entire out-of-sample evaluation period.

3.3. *Predictability over Time*

The recent literature emphasizes the instability of most macroeconomic predictor variables when predicting the equity risk premium (see, for example, Goyal and Welch (2008) and Paye and Timmermann (2006)). In contrast to these findings, Neely et al. (2014) show that technical indicators are less affected by this issue. Figure 1, therefore, plots the difference between the cumulative squared forecast error (CSFE) for the historical average benchmark (Equation (7)) and the CSFE for the trend-based forecasts (Equation (6)) for the set of risk factors considered in this study. These time series plots are informative graphical devices on the consistency of TI's out-of-sample predictive ability and allow the comparison of both competing forecasts for any particular evaluation period. More specifically, when the plotted curve increases, the trend-based forecast outperforms the historical average benchmark, while the opposite holds when the curve decreases.

[FIGURE 1 ABOUT HERE]

The plots in Figure 1 depict TI's superior, or at least equal, out-of-sample predictive ability over almost all prolonged evaluation periods from May 1969 to December 2015. Except for some subperiods, the trend-based forecasts consistently outperformed or performed at least as good as the historical average benchmark. Hence, TI's predictive ability documented in the previous subsections stems from the entire sample period rather than some special periods. In addition, the plot on the equity risk premium illustrates graphically the crash insurance characteristic of technical indicators, as the trend-based forecasts relative performance is strongest during NBER recessions and particularly during the height of the Global Financial Crisis in 2008 when equity prices dropped sharply.

In sum, this section documents significant time variation and predictability in a set of risk factors based on TI. This predictive ability is consistent over almost all prolonged evaluation periods and most pronounced during downtrends in the underlying risk factor. Hence, even factor risk premiums that seem to have vanished over time, may still be economically important in a conditional context. As technical indicators are primarily designed to detect trends in asset prices, these results imply substantial time series momentum in factor risk premiums.

4. Trend-based Dynamic Factor Strategy

Using the insights from the previous section, I introduce a trend-based dynamic factor strategy, in which the respective risk factor is levered up or down over time based on the

trend-based forecasts so as to maximize its unconditional Sharpe ratio. The design of the dynamic strategy is inspired by Daniel and Moskowitz (2016) who construct a dynamic momentum strategy using return forecasts based on an interaction between a bear market indicator and the market variance over the preceding six month and demonstrate that this strategy substantially outperforms the static momentum portfolio. In the Appendix of their paper, they show that, for the objective function of maximizing the unconditional Sharpe ratio in the context of an allocation decision between a risky and a risk free asset, the optimal weight on the risky asset i at time t can be calculated as

$$w_{i,t} = \left(\frac{1}{2\gamma} \right) \left(\frac{\hat{r}_{i,t+1}}{\hat{\delta}_{i,t+1}^2} \right), \quad (8)$$

where γ is a time-invariant risk adjustment coefficient that controls the unconditional risk and return of the dynamic strategy, $\hat{r}_{i,t+1}$ is the conditional expected return on risky asset i over period $t + 1$, and $\hat{\delta}_{i,t+1}^2$ is its conditional variance.⁷

In order to implement the trend-based dynamic strategy (denoted TRS) on risk factor j , I employ trend-based forecasts obtained from Equation (6) to provide an estimate of $\hat{r}_{j,t+1}$, the expected factor risk premium. In contrast to Daniel and Moskowitz (2016), who introduce two specific models to forecast both returns and variance of the momentum portfolio, my study only focuses on the forecastability of expected returns. Hence, I generate naïve variance forecasts using a five-year-moving window of past returns, which is the standard in the return predictability literature (see Neely et al. (2014), among many others). In addition, I set γ equal to 2.5 in the main analysis, which corresponds to a relative risk aversion coefficient of five for the mean-variance investor in the context of Campbell and Thompson (2008). My results, however, are also robust to alternative coefficients. Detailed results are provided in Table C3 in the Appendix. In addition, I follow the standard in the return predictability literature and constrain the optimal weights in the risky asset, w , to lie between 0 and 1.5, which imposes realistic portfolio constraints by precluding short sales and preventing more than 50% leverage. The dynamic strategy then realizes a return during period $t + 1$ equal to

$$r_{p,i,t+1} = w_{i,t} r_{i,t+1} + r_{f,t+1}, \quad (9)$$

where $r_{f,t+1}$ is the risk-free rate during period $t + 1$. In order to evaluate the trend-based dynamic factor strategy, I compare it to the static risk factor strategy (denoted BH), i.e. a

⁷Note that the optimal weights in Equation (8) are almost identical to those in Campbell and Thompson (2008), which are derived based on the decision of a mean-variance investor who allocates between a risky and a risk-free asset and are predominantly used in the return predictability literature in order to determine the economic value of a predictor variable. The only difference is that in Campbell and Thompson (2008) the term on the right-hand side of the equation is divided by γ instead of 2γ .

buy and hold strategy that always keeps a weight of one in the risky asset in Equation (9), and a dynamic strategy that uses out-of-sample return forecasts based on the historical average return (Equation (7)) to determine the optimal weights in Equation (8) (denoted HA). The performance measures I use for comparison are the certainty equivalent return (CER), Sharpe ratio, maximum drawdown (MD) and the annual turnover. Table 4 reports the results. The first, second and fourth column of Table 4 present the CER for the competing strategies, which is calculated as:

$$\text{CER}_{p,i} = \hat{\mu}_{p,i} - \frac{1}{2}\gamma\hat{\sigma}_{p,i}^2, \quad (10)$$

where $\hat{\mu}_{p,i}$ and $\hat{\sigma}_{p,i}^2$ are the mean and variance, respectively, for the strategy return on risk asset i over the evaluation period. Intuitively, the CER can be interpreted as the risk-free rate of return that an investor would be willing to accept instead of holding the risky portfolio. Hence, it can be interpreted as an annual portfolio management fee or transaction cost an investor would be willing to pay to have access to the trend-based dynamic factor strategy. The CER gain, presented in column three and five, is the difference between the CER for an investor who uses the trend-based dynamic factor strategy and the CER for an investor who uses either the historical average based strategy (column 3) or the static strategy (column 5). In order to assess whether the CER gain is statistically significant, I follow Demiguel et al. (2009) and apply asymptotic theory as in their paper.

Table 4 shows that the trend-based dynamic factor strategies generate substantial economic gains for investors, consistent with the previous results. Specifically, with the exception of WML, the annualized CER gains range from 1.46% to 5.26% and are statistically significant, regardless of whether the historical average based strategy or the static strategy is used as a benchmark. Similar results are obtained based on the strategies annualized Sharpe ratios reported in columns six to ten. In order to check whether the improvements in the risk-return profile are also statistically significant, I use the test introduced by Ledoit and Wolf (2008). Again these improvements are statistically and economically significant for almost all risk factors. Consider, for example, the HML portfolio. The Sharpe ratio increases from 0.49 for the historical average based dynamic strategy or 0.53 for the static strategy to 0.96 for the trend-based dynamic strategy.

[TABLE 4 ABOUT HERE]

With regard to tail risk characteristics, columns seven to nine report the maximum drawdown (MD) for the competing strategies. The MD is a popular measure of actively managed funds in practice, which is defined as the largest negative return from a price peak

to a bottom. Intuitively, the MD measures the maximum loss an investor incurred, if he had invested in the strategy at the worst possible time. All in all, the trend-based dynamic strategy substantially reduces the MD on all considered risk factors. Consider HML again, for example. The MD more than halves from 49.9% for the historical average based strategy and 42.3% for the static strategy to just 22.2%. This reduction is remarkable and particularly useful for investors who want to minimize their drawdown risk. Finally, to get a sense of the amount of trading required to implement the trend-based signals, the last two columns in Table 4 report the average annual turnover for the trend-based dynamic factor strategies as well as the historical average based strategies. The annual turnover for the trend-based dynamic strategy ranges between 3.7 and 6.4, which corresponds to the average holding days of the trend-based signals reported in Table 1.

To further underline the tail risk characteristics of the trend-based dynamic factor strategy, Table 5 shows the results for the popular market timing test of Treynor and Mazuy (1966) using the returns on the trend-based dynamic factor strategy applied on risk factor j , $r_{j,t}^{\text{TRS}}$, as the dependent variable:

$$r_{j,t}^{\text{TRS}} = \alpha_i + \beta_{r_j} r_{j,t} + \beta_{r_j^2} r_{j,t}^2 + \epsilon_{j,t}. \quad (11)$$

The market timing coefficients, $\beta_{r_j^2}$, are positive and statistically significant for all risk factors (except SMB). These results again highlight TI's crash insurance characteristic as the trend-based dynamic strategies deliver their highest returns during the most extreme market periods, i.e. the biggest up and down movements in the underlying risk factor. Hence, the strategies have payoffs similar to an option straddle on the underlying risk factor. The strategies generate this payoff structure because TI tends to issue a buy signal, indicating high future returns, when the risk factor is in a major uptrend and a sell signal, indicating low or even negative future returns, when the risk factor crashes.

[TABLE 5 ABOUT HERE]

To get a more intuitive understanding of this mechanism, Figure 2 compares the trend-based dynamic strategy to the static strategy during the biggest short-term drawdown observed for any risk factor during my sample period, the so-called "momentum crash" of 2009 (see Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015) for the recent literature on momentum crashes). There are two important observations to be made. First, as already noted in Daniel and Moskowitz (2016), crashes generally come in trends as they have relatively long duration and occur over the span of multiple months. Second, as technical indicators are primarily designed to detect trends, TI was able to prevent a large part of the momentum crash as it issued a sell signal just after the short and immediate trend reversal

at the outset of the crash. Hence, technical indicators prove particularly informative during crashes, because they often happen when the risk factor goes from normal to bad (leading to a sell signal of TI and a low weight in the risk factor within the trend-based dynamic strategy), and then from bad to worse (resulting in the large reduction in losses due to the low weight in the risk factor).

[FIGURE 2 ABOUT HERE]

The trend-based dynamic factor strategy can be interpreted as another hedged risk factor strategy comparable to the ones introduced in Grundy and Martin (2001), Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016). However, several points are worth mentioning. First, the aforementioned studies focus exclusively on WML, whereas my trend-based dynamic strategy is applicable on all risk factors and shows impressive results across a range of risk factors. This documented consistency provides some evidence for the robustness of my strategy and thereby guards against potential data mining. Second, the profitability of the trend-based dynamic could be further improved by relaxing some of the imposed restrictions. In contrast to Daniel and Moskowitz (2016), for example, I only focus on predicting risk factor returns and employ naïve variance forecasts. Using an additional sophisticated model to forecast the variance as in their study, would further improve the performance of my strategy. In addition, by following the standard in the return predictability literature, I restrict the optimal weights on the risk factor within the trend-based dynamic strategy between 0 and 1.5. Neglecting these constraints and allowing for short sales as well as the use of substantial leverage as in the aforementioned studies, would also improve the performance of the trend-based dynamic strategy, particularly during crashes. Third, my trend-based dynamic strategy is easily implementable in real time and less affected by potential measurement errors or model misspecification as it only depends on a binary buy or sell signal.

Note that an actual implementation of the trend-based dynamic strategy would certainly incur higher transaction costs than the historical average based benchmark strategy, even though potentially lower than those incurred by the dynamic strategy of Daniel and Moskowitz (2016). Nonetheless, as in Campbell and Thompson (2008) and Daniel and Moskowitz (2016), I abstract from any transaction costs in this study. This choice is motivated by at least three reasons. First, the impact of transaction costs on the profitability of risk factor portfolios varies considerably across risk factors and is still debated in the literature (see, for example, Novy-Marx and Velikov (2016)). Second, by comparing the annual turnover to the CER in Table 4, one can already get a first indication of the potential impact of transaction costs on the strategies' profitability, without having to specify them

explicitly. Third, to be in line with the literature on return predictability, I investigate a very naïve implementation of the trend-based signals, which was designed ignoring potential transaction costs and consider the most basic technical indicators. In practice, however, large institutional investment managers devote entire departments to the sole purpose of improving their investment signals and reducing transaction costs associated with their investment strategies.⁸ Fourth, Table B1 in the Appendix shows that my previous results also apply to both legs of the considered risk factor. Hence, long-only investors facing short-sale restrictions are also able to exploit TI's predictive ability. In addition, the results on the long leg are less affected by concerns regarding tradability, implementation and transaction costs. These points should certainly be taken into account when assessing practical implications of the trend-based dynamic factor strategies strong performance.

5. Trend-based Conditional Asset Pricing Models

The documented significant time variation and predictability in factor risk premiums suggests linking these findings to the cross-section of expected stock returns. Theoretically, it is well known that an asset pricing model could hold perfectly in a conditional framework, even though assets are mispriced in an unconditional context. Hence, an asset's conditional pricing error might be zero, even when its unconditional pricing error is not, if its factor loadings vary over time and are correlated with the respective factor risk premiums or factor volatilities. Consider the CAPM, for illustrative purposes. An asset with a conditional pricing error that is always zero under a given asset pricing model, might exhibit positive unconditional pricing errors, if it has a higher (lower) factor loadings particularly when exposure to the risk factor is rewarded with a higher (lower) risk premium, and vice versa for assets with negative unconditional pricing errors.

The aim of this section, therefore, is to examine whether TI's predictive ability has significant effects in conditional versions of the most popular asset pricing models: the CAPM of Sharpe (1964) and Lintner (1965), the three-factor model of Fama and French (1993) and the five-factor model of Fama and French (2014). To do so, I compare the aforementioned unconditional models to their conditional counterparts using the respective trend-based signals as conditioning variable within the framework of an asset pricing test. As test assets, I

⁸Novy-Marx and Velikov (2016) discuss several simple cost mitigation techniques for the implementation of risk factor portfolios. One potential technique would be to do the risk factor rebalancing on days when the trend-based signal changes. With regard to the implementation of trend-based signals, a simple way to reduce transaction costs would be to increase the lag length of the technical indicator. Table C1 in the Appendix provides results on the inherent relation between a technical indicator's lag length, profitability and annual turnover.

consider the 25 value-weighted size and momentum sorted portfolios and 30 industry-sorted portfolios from Kenneth French's Data Library. I focus on these portfolios because the prior literature has identified those as particular challenges for existing unconditional asset pricing models (see, among others, Fama and French (2015) for momentum-sorted portfolios and Fama and French (1997) for industry-sorted portfolios).

In order to compare the competing models, I start by estimating the prices of risk for the unconditional models using the Fama and MacBeth (1973) two-stage regression approach. In the first stage of the approach, the test assets' factor loadings have to be estimated by running the following time series regression separately for each test asset i :

$$r_{i,t+1} - r_{f,t+1} = \alpha_i + \beta_i^j r_{j,t+1} + \dots + \epsilon_{i,t+1}, \quad (12)$$

where $r_{i,t+1}$ is the return on test asset i and $r_{j,t+1}$ is the return on risk factor j . Subsequently, in the second stage, I investigate whether the test assets' average excess returns are related to their previously estimated factor loadings ($\hat{\beta}$) by running the following cross-sectional regression:

$$\overline{r_i - r_f} = \gamma_0 + \gamma_j \hat{\beta}_i^j + \dots + \epsilon_i, \quad (13)$$

where $\overline{r_i - r_f}$ is the average excess return on test asset i . If the test assets' factor loadings are important determinants of their average returns, the estimated prices of risk, γ_j should be significantly greater than zero. In addition, theory says that the regression intercept (γ_0), i.e. zero-beta rate, should be zero if the model is correctly specified.

In contrast to their unconditional counterparts, conditional models allow for time variation in the factor loadings. A common approach to empirically investigate conditional asset pricing models, is to model factor loadings as a linear function of observed conditioning variables that are assumed to be related to the factor risk premiums (Jagannathan and Wang (1996)). Hence, as the previous sections document a high (low) factor risk premium following uptrends (downtrends) in the underlying risk factor as indicated by buy (sell) signals of TI, I model the conditional factor loading of test asset i on risk factor j as

$$\beta_{i,t}^j = \beta_{i,Down}^j + \beta_{i,Up-Diff}^j \text{TI}_{j,t} \quad \text{for all considered risk factors } j. \quad (14)$$

Intuitively, $\beta_{i,Down}^j$ ($\beta_{i,Up}^j = \beta_{i,Down}^j + \beta_{i,Up-Diff}^j$) can be interpreted as the loading of test asset i on risk factor j following downtrends (uptrends) in the risk factor, i.e. when the factor risk premium is low (high). Replacing the constant factor loadings in Equations (12) and (13) with their time-varying counterparts then yields a trend-based conditional asset pricing model:

$$r_{i,t+1} - r_{f,t+1} = \alpha_i + (\beta_{i,Down}^j + \beta_{i,Up-Diff}^j \text{TI}_{j,t}) r_{j,t+1} + \dots, \quad (15)$$

which can also be rearranged to an unconditional model:

$$r_{i,t+1} - r_{f,t+1} = \alpha_i + \beta_{i,Down}^j r_{j,t+1} + \beta_{i,Up-Diff}^j \text{TI}_{j,t} r_{j,t+1} + \dots \quad (16)$$

Note that Equation (16) is equivalent to Equation (12) with additional scaled risk factors.⁹ If in fact, the conditioning information is important for explaining the cross-section of expected stock returns, the regression coefficients on the trend-based signals, $\beta_{i,Up-Diff}^j$, should be positive and statistically significant.

Panel A in Table 6 reports results of asset pricing tests using the 25 size and momentum sorted portfolios as test assets. In addition to the annualized prices of risk and their respective t -statistics using Shanken (1992)-corrected standard errors, the adjusted R^2 statistic of the second-stage cross-sectional regression and the square root of the annualized mean squared pricing error (PE) across all test assets are reported. The first three rows confirm the well-known failure of the aforementioned unconditional asset pricing models at explaining momentum-sorted portfolios (see Fama and French (2015)). The regression intercepts are strongly statistically significant and economically large. In addition, the prices of risk for RMRF, HML and CMA are negative and strongly statistically significant.

[TABLE 6 ABOUT HERE]

However, allowing for time variation in the factor loadings substantially improves the explanatory power of the aforementioned asset pricing models. Specifically, with regard to the trend-based CAPM, row 4 documents an economically substantial increase in the cross-sectional adjusted R^2 from 23.1% to 60.3%, which is also highly statistically significant based on the R^2 comparison test of Kan et al. (2013). Moreover, the price of risk of the additional scaled RMRF factor is positive and strongly statistically significant and the PE reduces from 3.46% to 2.43% per annum. To get a more intuitive understanding of this result, Table 7 reports the average return (Panel A), unconditional market loading (Panel B), conditional market loading following uptrends in RMRF (Panel C), and conditional market loadings following downtrends in RMRF (Panel D) for each test asset. Panel A and B confirm the prior finding in the literature that the unconditional CAPM is unable to explain momentum-sorted portfolio. Winner portfolios, even though they yield higher

⁹Equation (16) assumes that the alphas are constant over time. This assumption is motivated by at least two reasons. First, the literature documents a negligible effect of time varying alphas in the context of conditional asset pricing models (Lettau and Ludvigson (2001)). And second, the choice of the relevant conditioning variable for each alpha is not clear if different conditioning variables are used for each risk factor, as in my case.

average returns over time than loser portfolios, exhibit lower unconditional market loadings. However, Panel C and D reveal that once I allow for time variation in the market loadings, it appears that the winner portfolios' conditional market loadings are generally higher (lower) following uptrends (downtrends) in RMRF when the equity risk premium is high (low).

[TABLE 7 ABOUT HERE]

This pattern is of course well known in the literature. Kothari and Shanken (1992) argue that, by their nature, past-return sorted portfolios have significant time-varying loadings to systematic risk. Specifically, because winner (loser) portfolios realized high (low) recent returns, they have positive (negative) loadings on factors that have had a positive realization, and negative (positive) loadings on factors which have had negative realizations, during their formation period. However, as outlined earlier, time variation in the factor loadings is not enough to improve unconditional asset pricing models, instead the factor loadings need to be high (low) precisely when exposure to the factor is rewarded with a high (low) risk premium. So far, the prior literature that tries to explain momentum-sorted portfolios with time varying exposure to systematic risk, such as Grundy and Martin (2001) and Daniel and Moskowitz (2016), fail to establish the necessary correlation between factor loadings and factor risk premiums. Hence, they conclude that controlling for time-varying exposures to common risk factors does not affect the momentum anomaly. Specifically, considering a look-back period, i.e. lag length, over the last seven months (excluding the last month), Grundy and Martin (2001) find an insignificant (or even negative) time series momentum effect in the FF3 factors (see their Equation (8)). Similarly, Daniel and Moskowitz (2016) find that a conditional version of the FF3 that allows for a different market loadings of the momentum portfolio's market loading during crash states (defined based on the past two-year market return) does not reduce, but instead increase its abnormal returns.

Similar improvements can be observed for the FF3 and FF5. Their explanatory power improves substantially after taking account of the time variation in factor loadings. The cross-sectional adjusted R^2 increases from 61.3% to 84.1% for the FF3 and from 66.4% to 83% for the FF5. Again these improvements are statistically significant at the 1% level based on the R^2 comparison test of Kan et al. (2013). Even more strikingly, the zero-beta rate, which is economically large and strongly statistically significant for the unconditional models, becomes statistically insignificant for both trend-based conditional counterparts. Similarly, the annualized PE reduces substantially from 2.35% to 1.39% for the FF3 and from 2.08% to 1.27% for the FF5. As for the CAPM, the strong results are driven by the positive correlation between the factor loadings and the respective factor risk premiums.

Panel B of Table 6 documents similar results using the 30 industry-sorted portfolios as

test assets. Rows seven through nine highlight the well-known failure of the unconditional models in explain industry returns (Fama and French (1997)). However, allowing for time variation in the factor loadings again substantially improves the explanatory power of the considered asset pricing models. The cross-sectional adjusted R^2 increases from 1.6% to 45.6% (CAPM), from 14.3% to 44.2% (FF3), and from 30.2% to 47% (FF5), while the PE reduces from 1.74% to 1.27% (CAPM), from 1.57% to 1.19% (FF3), and from 1.36% to 1.05% (FF5). Again the improvements in explanatory power are strongly statistically significant. The substantial increase in explanatory power for the trend-based conditional CAPM is particularly remarkable as its cross-sectional adjusted R^2 even exceeds that of the unconditional FF3 and FF5.

Note that each trend-based conditional model (Equation (16)) nests its unconditional counterpart (Equation (12)), in which the regression coefficients on the scaled factors, $\beta_{i,Up-Diff}^j$, are zero. Hence, to further support TI's significant effects within the trend-based conditional models, I test, for each test asset i separately, the joint null hypothesis,

$$\beta_{i,Up-Diff}^j = 0 \quad \text{for all risk factors } j. \quad (17)$$

This exclusion test essentially examines the joint significance of the trend-based signals as a group. Using heteroskedasticity-robust χ^2 -statistics, the restrictions in Equation (17) are strongly rejected for all 25 size and momentum sorted portfolios within all three considered asset pricing model. With respect to the 30 industry-sorted portfolios, the exclusion restrictions are rejected for all but seven (CAPM), three (FF3), and two (FF5) industry portfolios. The complete list of χ^2 -statistics is reported in Table D1 in the Appendix.

All in all, this section documents that accounting for the time variation in factor loadings substantially improves the ability of existing asset pricing models to explain the cross-section of expected stock returns.¹⁰ These results provide additional evidence that technical indicators represent genuine economic information and highlight the importance of accounting for time series momentum in factor risk premiums when determining an asset's expected return.

¹⁰Note that the aim of this section is not to provide a fully-fledged quantitative test of conditional asset pricing models. Instead, I want to emphasize the importance of accounting for time series momentum in factor risk premiums when determining an asset's expected return. Full conditional models are not testable empirically, as they imply a conditional linear factor model with respect to the investors full information set, which is not observable. Moreover, Lewellen and Nagel (2006) argue that several restrictions would need to be imposed on the cross-sectional coefficient estimates in Equation (16) for a full quantitative test of a conditional asset pricing model. However, fulfilling these two criteria is beyond the scope of this paper.

6. Economic Explanation

Overall, my results pose several challenges to risk-based asset pricing theories. Two main points are worth emphasizing. First, risk-based explanations typically interpret time-varying factor risk premiums as a reflection of a time-varying compensation for bearing aggregate risk. Specifically, factor risk premiums should be higher (lower) during bad (good) economic times when aggregate risk aversion, aggregate risk, and consequently also marginal utility are high (low) (Fama (1991)). In Section 3, however, I document significant higher (lower) factor risk premiums following recent uptrends (downtrends) in the underlying risk factor. If one makes the straightforward assumption that recent prolonged uptrends (downtrends) in a risk factor are due to positive (negative) news and hence represent improving (deteriorating) economic times, this finding appears inconsistent with the aforementioned risk-based argument.¹¹ Supporting the inconsistency of the documented predictability with time-varying compensation for aggregate risk, Table A1 in the Appendix shows low and diverse pairwise correlations between the factor risk premiums as implied by their respective trend-based signals as well as the returns on the trend-based dynamic factor strategies. As a result, the findings of Section 5 that the positive unconditional pricing errors of certain test assets can be explained by higher (lower) factor loadings following uptrends (downtrends) in the respective risk factors, when the factor risk premiums are particularly high (low), (and vice versa for test assets with negative unconditional pricing errors), also seems inconsistent with a risk-based explanation. And second, proponents of market efficiency generally argue that risk factors exhibit substantial positive risk premiums, because they tend to perform badly exactly when investors need it the most (Fama and French (1993)). However, Section 4 reveals substantial improvements in the risk factors' risk-return profile as well as reductions in their drawdowns using the trend-based dynamic strategy. These results also question risk-based asset pricing theories.

But what explains these findings? In the following subsections, I investigate several potential explanations and sources of predictability. These are related to rational theories of momentum, volatility risk, liquidity risk, investor sentiment, and information uncertainty.

6.1. *Rational Theories of Momentum*

Rational explanations of momentum starting with Berk et al. (1999) and Johnson (2002) state that firms that performed well (bad) in the recent past, i.e. experienced a recent

¹¹Huang et al. (2016), for example, introduce a market state indicator based on a 200-day moving average, which either indicates good economic times if the level of the S&P 500 is above its 200-day MA or bad economic times otherwise.

uptrend (downtrend) in their stock price, are better (worse) poised to exploit existing growth options. Consequently, as those growth options are equivalent to a risky asset that accounts for a large fraction of firm value, such firms are also more risky and, in turn, associated with higher expected returns. As argued in Sagi and Seasholes (2007), these theories imply that growth firms (short leg of HML) and firms with high margins (long leg of RMW) should exhibit stronger time series momentum compared to value firms (long leg of HML) and firms with low margins (short leg of RMW), respectively. My empirical results documented in Table B1 in the Appendix, however, reveal the opposite. Moreover, these theories cannot explain the substantially stronger time series momentum observed in small firms (long leg of SMB) compared to large firms (short leg of SMB). Hence, explanations related to rational theories of momentum appear unlikely to explain my previous findings.

6.2. *Volatility and Liquidity Risk*

In two influential studies, Merton (1980) and French et al. (1987) argue that lower stock market volatility implies lower market risk and consequently also a lower risk premium. Hence, in order to investigate whether TI's predictive ability is due to the fact that it proxies for time variation in factor risk as measured by factor volatility, I follow the aforementioned literature and estimate the following predictive regression for each risk factor j :

$$LVOL_{t,j} = \alpha_j + \beta_j \overline{\text{TI}}_{t,j} + \psi_j LVOL_{t-1,j} + \epsilon_{t,j}, \quad (18)$$

where $LVOL_{t,j} = \log(\sqrt{FVAR_{t,j}})$ is the monthly aggregate volatility and $\overline{\text{TI}}_{t,j}$ is the average trend-based signal, both during month t . The monthly aggregate factor variance, $FVAR_{t,j}$, is the sum of squared daily returns on risk factor j within month t calculated as

$$FVAR_{t,j} = \sum_{d=1}^{N_t} r_{j,d}^2, \quad (19)$$

where N_t is the number of trading days within month t , and $r_{j,d}$ is the daily return on risk factor j on the d^{th} trading day of month t . Following the literature, I also include lagged values of aggregate volatility as volatility is highly persistent.

Given that factor risk premiums are higher following uptrends ($\text{TI} = 1$) in the underlying risk factor, the volatility-based argument implies a higher factor volatility and thus factor risk during these periods. In terms of Equation (18), we would expect the slope coefficient on $\overline{\text{TI}}_{t,i}$, β_i , to be positive. However, Panel A of Table 8 reveals a statistically significant negative relation instead (with the exception of WML), inconsistent with the volatility-based explanation. Likewise, I find in unreported results that, if different, volatility is generally

lower (higher) during uptrends (downtrends) in the underlying risk factor when the factor risk premium is particularly high (low). Hence, it seems unlikely that factor risk as measured by factor volatility is driving TI's strong predictive ability.

[TABLE 8 ABOUT HERE]

Using the same framework, I also test whether TI's predictive ability is partly driven or exaggerated by aggregate liquidity risk. As negative liquidity shocks result in low contemporaneous returns and high expected returns, we would expect a negative relation between liquidity and the trend-based signals. Using the aggregate liquidity measure of Pastor and Stambaugh (2003), the results in Panel B of Table 8, however, suggest no coherent relationship. If anything, I find a positive relationship for $\overline{\text{TI}}_{RMRF}$ and $\overline{\text{TI}}_{SMB}$, inconsistent with a liquidity-based explanation.

6.3. *Sentiment and Informational Uncertainty*

The behavioral finance literature starting with Daniel et al. (1998), Barberis et al. (1998) and Hong and Stein (1999) demonstrate that financial markets are informationally inefficient as investors', due to a plethora of behavioral biases, initially underreact and subsequently overreact to new information and thereby create time series momentum in asset prices. Due to limits to arbitrage, these deviations from fundamental values then might persist, even if rational investors recognize the mispricing (Shleifer and Vishny (1997)). The initial under- and subsequent overreaction can then be further intensified by investor sentiment and informational uncertainty. Baker and Wurgler (2006, 2007) argue that investors underreact even more to good (bad) news when sentiment is pessimistic (optimistic) and overreact even more to good (bad) news when investor sentiment is optimistic (pessimistic). Somewhat related, Zhang (2006) shows that if investors underreact to new information, they will underreact even more in case of greater information uncertainty. In addition, behavioral biases, which are the fundamental driving force of investors' under- and overreaction to new information, are even more pronounced when there is more uncertainty (Hirshleifer (2001)).

With regards to effects on the cross-section of stock returns, Baker and Wurgler (2006) document that stocks that are more difficult to value, i.e. small (long leg of SMB), growth (short leg of HML), unprofitable (short leg of RMW) or high investment (short leg of CMA) stocks, are more affected by investor sentiment, as they are more sensitive to speculative demands and more costly to arbitrage.¹² By definition, these stocks that are more difficult

¹²The evidence on the effect of investor sentiment on WML is mixed: Baker and Wurgler (2006) report no statistically significant relation, while Antoniou et al. (2013) document a positive effect.

to value are also more affected by information uncertainty (see also Zhang (2006)). As a result, these stocks should be more affected by investors' initial under- and subsequent overreaction to new information and therefore exhibit stronger time series momentum.

Hence, if part of TI's predictive ability and consequently the existence of time series momentum in factor risk premiums is related to theories of sentiment and an informationally inefficient market, I should make the following three observations.

First, TI's predictive ability should be stronger on stocks that are more difficult to value. The results documented in Table B1 in the Appendix, support this hypothesis, as predictability is generally stronger for small, unprofitable and high investment stocks. The hypothesis is also support by previous findings in the literature. Han et al. (2013) show that technical indicators perform particularly well on stocks that are more affected by information uncertainty. From a theoretical point of view, Zhu and Zhou (2009) illustrate that technical indicators are a valuable learning tool if the market dynamics are uncertain.

Second, TI should indicate an uptrend (downtrend) on RMRF and SMB as well as a downtrend (uptrend) on HML, RMW and CMA during periods of increasing (decreasing) investor sentiment. Using the Baker and Wurgler (2006) orthogonalized sentiment-change index ($\Delta\text{SENT}^\perp$), I investigate this second hypothesis by running the following bivariate logit regression separately for each risk factor j :¹³

$$\overline{\text{TI}}_{t,j} = \alpha_j + \beta_j \Delta\text{SENT}_t^\perp + \epsilon_{t,j}, \quad (20)$$

where $\overline{\text{TI}}_{t,j}$ is the average trend-based signal on risk factor j during month t .¹⁴ The results reported in Panel A of Table 9 confirm the aforementioned hypothesis. Supporting this result, Neely et al. (2014) find that part of technical indicators' ability to predict the equity risk premium stems from their ability to predict changes in sentiment, which are both sizably contemporaneously correlated.

Third, TI should indicate a downtrend (uptrend) on RMRF and SMB as well as an uptrend (downtrend) on HML, RMW and CMA after high (low) levels of investor sentiment, when sentiment-induced mispricing is likely to get corrected. I investigate the last hypothesis using the following bivariate logit regression separately for each risk factor j :

$$\overline{\text{TI}}_{t,j} = \alpha_j + \beta_j \text{SENT}_{t-1}^\perp + \epsilon_{t,j}. \quad (21)$$

¹³I consider the orthogonalized sentiment-change index in order to ensure that the index is not related to systematic risk.

¹⁴I consider the average trend-based signal during month t instead of the signal at the end of month $t - 1$ as usually done in predictive regressions, because the trend-based signals are calculated on a daily basis while the investor sentiment index is only updated monthly. Using the trend-based signal at the end of month $t - 1$ could bias my results, if the signal changes during month t .

Again the results reported in Panel B of Table 9 confirm the aforementioned hypothesis.

[TABLE 9 ABOUT HERE]

Overall, the evidence presented in this section suggests that TI's superior predictive ability is unlikely explained by risk-based asset pricing theories. In contrast, my findings appear to be more consistent with theories of sentiment and an informationally inefficient market in which investor's initial under- and subsequent overreaction to new information leads to time series momentum in factor risk premiums.

But, even though my results challenge the random walk hypothesis and perhaps also a more sophisticated notion of market efficiency with time-varying risk premiums reflecting the time-varying compensation for bearing aggregate risk, I cannot rule out the existence of a rational theory that can explain my findings. Nonetheless, any potential risk-based asset pricing theory also has to explain my empirical results.

7. Conclusion

In this paper, I document significant time variation and predictability in a set of risk factors based on technical indicators. As these indicators are primarily designed to detect trends in asset prices, these findings imply substantial time series momentum in factor risk premiums. Specifically, factor risk premiums are significantly larger (lower) following recent uptrends (downtrends) in the underlying risk factor. The documented predictive ability can be utilized in several areas of asset pricing, such as improving the risk-return profile of static risk factors as well as the explanatory power of empirical asset pricing models. On a more general level, technical indicators can improve the estimation of an asset's expected return.

My evidence appears inconsistent with risk-based asset pricing theories as the predictive ability is particularly strong during factor crashes and unlikely explained by factor risk, liquidity risk, rational theories of momentum, or an aggregate state variable. In addition, if one makes the straightforward assumption, that recent uptrends (downtrends) in a given risk factor are due to good (bad) recent news, the finding of high (low) factor risk premiums following uptrends (downtrends) also seems to contradict conventional risk-based explanations proposed for the time variation and predictability in factor risk premiums. In contrast, my evidence appears more consistent with theories of sentiment and an informational inefficient market in which investors' initial under- and subsequent overreaction to new information leads to time series momentum in factor risk premiums. Overall, my findings present new evidence and pose several challenges for existing risk-based asset pricing theories and future research.

Appendix A. Correlations

[TABLE A1 ABOUT HERE]

Appendix B. Long Leg and Short Leg Analysis

Each risk factor considered in this study consists of two subportfolios: a long leg covering the higher-performing stocks that are bought and a short leg covering the underperforming stocks that are sold short. In this subsection, I therefore run my predictive regression analysis (see Section 3) as well as the trend-based dynamic strategy (see Section 4) separately on both legs of each considered risk factor. This analysis is interesting for at least four reasons. First, it allows me to investigate whether long-only investors that face short-sale restrictions can also exploit my findings. Second, the results on the long leg are less affected by concerns regarding tradability, implementation and transaction costs. Third, it serves as an additional robustness check of my previous results. And fourth, it provides first insights into the potential sources of predictability (see Section 6).

[TABLE B1 ABOUT HERE]

Appendix C. Robustness Checks

Table C1 reports results of the predictive regression analysis (see Section 3) and trend-based dynamic strategies (see Section 4) using the alternative parameterizations and technical indicators introduced in Section 2.2. In addition, the last two columns in Table C1 present results using the first and second principal component extracted from the entire set of individual technical indicators. Combination methods are informative as they filter out much of the noise inherent in individual predictor variables, thereby also guarding against in-sample overfitting. The results highlight the inherent trade-off between profitability and annual turnover for different parameterizations, which I already discussed in Section 2.2. Specifically, a longer (shorter) lag length on the one hand decreases (increases) the annual turnover, but on the other hand also decreases (increases) the indicator's profitability. Overall, however, my results are robust to alternative parameterizations and technical indicators.¹⁵

¹⁵Note that the *MSFE-adjusted* statistics indicate that the MSFEs for some trend-based forecasts on RMRF and SMB are significantly less than that of the historical average benchmark, despite a negative R^2_{OOS} . Even though this result appears strange, it is possible when comparing nested model forecasts (Neely

[TABLE C1 ABOUT HERE]

So far, I only consider individual return forecasts, which depend on one of the nine technical indicator introduced in Section 2.2 (see Equation (6)). Hence, as an additional robustness check, Table C2 reports results using five combination methods proposed by Rapach et al. (2010). These methods combine the individual return forecasts and are therefore less reliant on one specific indicator or parameterization. Overall, the results are even stronger than using individual forecast.

[TABLE C2 ABOUT HERE]

Finally, Table C3 reports results using alternative coefficients for the time-invariant scalar, γ , which was used in Section 4 to control the unconditional risk and return of the trend-based dynamic factor strategy. All in all, my results are also robust to alternative coefficients.

[TABLE C3 ABOUT HERE]

Appendix D. Exclusion Tests for the Trend-based Conditional Asset Pricing Models

[TABLE D1 ABOUT HERE]

et al. (2014), Clark and West (2007), and McCracken (2007)). As also noted in Neely et al. (2014), under the null hypothesis that the constant expected return model generates the data ($\beta = 0$ in Equation (4)), the trend-based forecast in Equation (6) are noisier than the historical average forecast in Equation (7) because it estimates slope coefficients, β , with zero population value. Consequently, we expected the historical average MSFE to be smaller than the trend-based MSFE under the null. The *MSFE-adjusted* statistic accounts for negative expected difference under the null, so that the null can also be rejected, even if the R_{OS}^2 is negative

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Table 1: **Summary Statistics.** This table presents summary statistics for the set of risk factors and the respective trend-based signals. I consider the market (RMRF), size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum (WML) factors. Panel A reports, for each risk factor separately, its annualized mean return (Mean), annualized standard deviation (Std. Dev.), Sharpe ratio calculated as the average return divided by its standard deviation, lowest daily return observed over the entire sample period (Min), largest daily return (Max), and first order autocorrelation. Panel B reports the mean (Mean), standard deviation (Std. Dev.), average number of days until a change in the trend-based signal occurs (Avg. Holding Days), and first order autocorrelation of the respective trend-based signals. The sample period is from July 1963 to December 2015.

	RMRF	SMB	HML	RMW	CMA	WML
Panel A: Risk Factor Returns						
Mean (%)	4.85	1.85	3.94	3.04	3.54	7.81
Std. Dev (%)	15.68	8.22	7.84	5.49	5.76	11.05
Sharpe Ratio	0.31	0.23	0.50	0.55	0.61	0.71
Min (%)	-17.44	-11.19	-5.12	-2.86	-5.90	-8.22
Max (%)	11.35	6.10	4.01	4.06	2.51	7.05
Autocorrelation	0.05	0.05	0.15	0.16	0.15	0.22
Panel B: Trend-based Signals						
Mean	0.65	0.52	0.58	0.64	0.61	0.72
Std. Dev	0.48	0.50	0.49	0.48	0.49	0.45
Avg. Holding Days	72.76	60.43	67.86	69.85	77.51	100.14
Autocorrelation	0.94	0.95	0.95	0.94	0.95	0.95

Table 2: **In-Sample Predictive Regression.** This table presents ordinary least squares estimates for the bivariate predictive regression model outlined in Equation (4):

$$r_{i,t+1} = \alpha_i + \beta_i \text{TI}_{i,t} + \epsilon_{i,t+1},$$

where $r_{i,t+1}$ is the return on asset i for period $t + 1$ and $\text{TI}_{i,t}$ is the trend-based signal available at the end of period t . Reported are the regression coefficients (in percentages) and the associated heteroskedasticity-consistent t -statistics. p -values are based on a one-sided (upper-tail) wild bootstrap. The R^2 statistic is reported for the full sample period. The R_{EXP}^2 (R_{REC}^2) statistics are computed for National Bureau of Economic Research-dated business-cycle expansions (recessions), while R_{UP}^2 (R_{DOWN}^2) statistics are computed for up-trends (downtrends) in the underlying risk factor, defined based on whether the factor return during the previous year was positive (negative). All R^2 statistics are in percentages. IVX is the IVX-Wald statistic introduced by Kostakis et al. (2015). *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from April 1964 to December 2015.

	OLS							IVX
	α (%)	β (%)	R^2	R_{EXP}^2	R_{REC}^2	R_{UP}^2	R_{DOWN}^2	
RMRF	-0.25 (-0.05)	9.93** (1.78)	0.03	-0.03	0.04	-0.03	0.07	4.85**
SMB	-2.33 (-1.38)	9.42*** (3.89)	0.12	0.17	-0.09	0.11	0.07	16.34***
HML	-3.43** (-2.07)	14.03*** (5.91)	0.27	0.37	0.01	0.10	0.29	32.94***
RMW	-1.66 (-1.23)	7.86*** (4.59)	0.17	0.16	0.18	0.04	0.14	23.91***
CMA	-0.99 (-0.90)	8.02*** (4.90)	0.17	0.13	0.01	0.11	0.13	21.35***
WML	-0.04 (-0.01)	12.19*** (3.25)	0.09	0.12	-0.06	0.04	0.17	10.55***

Table 3: **Out-of-Sample Forecasting.** This table reports the out-of-sample performance of TI in predicting risk factor returns. All forecasts are estimated recursively using the data available at the end of the forecast formation period t . R_{OOS}^2 are the Campbell and Thompson (2008) out-of-sample R^2 statistics (in percentages), measuring the proportional reduction the the MSFE for trend-based forecasts relative to the historical average benchmark. CW-test is the Clark and West (2007) *MSFE-adjusted* statistic and DM-test is the modified Diebold and Mariano (1995) t -statistic. Both statistics test the null hypothesis that the historical average forecast MSFE is less than or equal to the competing trend-based forecast MSFE against the one-sided (upper-tail) alternative hypothesis. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The R_{OOS}^2 statistics are also reported separately for uptrends (Up) and downtrends (Down) in the underlying risk factor, defined based on whether the factor return during the previous year was positive or not. The e^2 term and Rem. term are the squared forecast bias and the remainder term, respectively, for the Theil (1971) MSFE decomposition. Both terms are reported for forecasts based on either TI or the historical average return (HA). The sample period is from May 1969 to December 2015.

	R_{OOS}^2	CW-test	DM-test	R_{OOS}^2		e^2		Rem. Term	
				Up	Down	TI_i	HA_i	TI_i	HA_i
RMRF	0.01	1.66**	0.15	-0.06	0.07	0.004	0.004	105.47	105.48
SMB	0.04	3.94***	0.34*	-0.07	0.17	0.005	0.009	28.77	28.78
HML	0.25	5.25***	2.75***	0.13	0.41	0.000	0.002	26.25	26.32
RMW	0.16	3.97***	2.14***	0.16	0.16	0.003	0.004	12.70	12.72
CMA	0.15	4.73***	2.03***	0.14	0.18	0.000	0.000	13.63	13.66
WML	0.07	2.89***	1.38***	0.03	0.25	0.001	0.002	51.22	51.25

Table 4: **Performance of the Trend-based Dynamic Factor Strategy.** This table compares the performance of three competing risk factor strategies. TRS is the trend-based dynamic factor strategy, for which the weights in the risk factor are determined based on Equation (8) using out-of-sample trend-based forecasts implied by Equation (6). HA is the historical average based factor strategy, for which the weights in the risk factor are also determined based on Equation (8) but using out-of-sample historical average forecasts implied by Equation (7). Finally, BH is the static factor strategy, which is equivalent to a buy-and-hold strategy on the risk factor. The respective out-of-sample forecasts are estimated recursively, using only data available at the end of the portfolio formation period t . CER is the annualized certainty equivalent return for the investor (in percentages). The Sharpe ratio is annualized and calculated as the mean portfolio return in excess of the risk-free rate divided by its standard deviation. MD is the maximum drawdown defined as the largest negative return over the sample period from a peak to a bottom (in percentages). Turnover is the average number of times wealth is traded each year. Δ denotes the difference in the respective performance measure between the trend-based dynamic factor strategy and either the historical average based factor strategy or the static factor strategy. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from May 1969 to December 2015.

	CER (%)			Sharpe Ratio			MD (%)			Turnover	
	TRS _{<i>j</i>}	Δ HA _{<i>j</i>}	Δ BH _{<i>j</i>}	TRS _{<i>j</i>}	Δ HA _{<i>j</i>}	Δ BH _{<i>j</i>}	TRS _{<i>j</i>}	HA _{<i>j</i>}	BH _{<i>j</i>}	TRS _{<i>j</i>}	HA _{<i>j</i>}
RMRF	8.30	3.02***	4.23**	0.56	0.34***	0.21	20.8	30.9	54.7	4.3	0.6
SMB	6.48	3.06***	2.27**	0.38	0.38**	0.25**	34.1	47.2	34.0	6.4	0.5
HML	12.15	4.66***	4.36***	0.96	0.47***	0.43***	22.2	49.9	42.3	5.7	0.1
RMW	10.08	2.48***	2.29***	0.95	0.38***	0.34***	16.0	45.0	34.5	6.2	0.7
CMA	10.39	1.75***	1.83***	0.88	0.23**	0.16*	19.8	26.0	17.8	5.5	0.4
WML	13.15	-0.08	3.08***	0.91	0.01	0.20*	42.5	46.6	63.6	3.7	0.1

Table 5: **Crash Characteristics of the Trend-based Dynamic Factor Strategy.** This table presents the ordinary least squares estimates of the Treynor and Mazuy (1966) market timing regression outlined in Equation (11). Both alphas and R^2 statistics are in percentages. In addition, the alphas are annualized. Newey and West (1987) robust t -statistics are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from May 1969 to December 2015.

	α (%)	β_R	β_{R^2}	R^2
RMRF	5.76 (4.31)	0.44 (10.32)	0.95 (2.08)	39.8
SMB	6.45 (4.78)	0.84 (10.79)	0.82 (0.65)	61.0
HML	5.60 (6.49)	0.86 (14.38)	4.15 (4.37)	69.9
RMW	6.28 (8.31)	0.72 (8.97)	3.08 (2.57)	62.3
CMA	5.08 (8.44)	0.96 (14.83)	4.26 (2.99)	74.6
WML	3.57 (2.40)	1.07 (18.48)	2.01 (2.65)	79.9

Table 6: Asset Pricing Tests. This table reports the ordinary least squares estimates for the second-stage cross-sectional regression of the Fama and MacBeth (1973) approach. In the first-stage, the factor loadings are computed in one time series regression of the test asset returns on the respective risk factors. The scaling variables are the trend-based signals calculated on the respective risk factor. As test assets, I use either 25 size and momentum sorted portfolios (Panel A) or 30 industry-sorted portfolios (Panel B). The regression coefficients are annualized and Shanken (1992) adjusted t -statistics are in parentheses. R^2 is the adjusted cross-sectional R^2 statistic of the second-stage cross-sectional regression. PE is the annualized square root of the mean squared pricing errors across all test assets. Both measures are in percentages. The sample period is from April 1964 to December 2015.

	Int.	Factors					Scaled Factors					R^2	PE
		RMRF	SMB	HML	RMW	CMA	RMRF	SMB	HML	RMW	CMA		
Panel A: 25 Size and Momentum Sorted Portfolios													
1	0.26 (8.13)	-0.14 (-3.98)										23.1	3.46
2	0.28 (7.11)	-0.14 (-3.90)	0.02 (1.43)	-0.11 (-4.15)								61.3	2.35
3	0.34 (3.14)	-0.20 (-2.12)	0.04 (2.63)	-0.18 (-3.23)	0.15 (2.33)	-0.20 (-2.42)						66.4	2.08
4	0.20 (7.36)	-0.07 (-2.03)					0.14 (2.63)					60.3	2.43
5	0.07 (1.17)	0.05 (0.97)	0.04 (3.35)	0.17 (2.68)			-0.27 (-1.32)	0.21 (1.07)	0.33 (5.62)			84.1	1.39
6	0.15 (1.23)	-0.03 (-0.31)	0.05 (2.70)	0.18 (1.23)	-0.05 (-0.55)	0.05 (0.40)	-0.26 (-0.94)	0.30 (1.85)	0.26 (2.95)	0.08 (1.14)	0.03 (0.23)	83.0	1.27
Panel B: 30 Industry-sorted Portfolios													
7	0.14 (5.58)	-0.02 (-0.72)										1.6	1.74
8	0.14 (6.40)	-0.02 (-0.63)	-0.01 (-0.52)	-0.02 (-1.40)								14.3	1.57
9	0.11 (5.72)	0.01 (0.28)	-0.01 (-0.37)	-0.03 (-1.95)	0.04 (2.38)	-0.02 (-0.93)						30.2	1.36
10	0.10 (7.18)	0.02 (0.85)					0.18 (4.43)					45.6	1.27
11	0.09 (3.72)	0.03 (0.94)	0.01 (0.71)	0.00 (-0.24)			0.21 (3.61)	0.06 (1.20)	0.05 (1.87)			44.2	1.19
12	0.10 (4.51)	0.02 (0.65)	0.00 (0.15)	0.01 (0.30)	0.02 (1.51)	-0.03 (-1.64)	0.21 (2.58)	0.07 (1.44)	0.07 (2.83)	0.01 (0.54)	0.02 (0.62)	47.0	1.05

Table 7: **Average Returns and Factor Loadings.** This table reports, for the 5 size and momentum sorted portfolios, their average annualized returns (Panel A) (in percentages), unconditional market loadings, β^{RMRF} , (Panel B), conditional market loading following downtrends, β_{Down}^{RMRF} , (Panel C), and the conditional market loading following uptrends, $\beta_{Up}^{RMRF} = \beta_{Down}^{RMRF} + \beta_{Up-Diff}^{RMRF}$, (Panel D). The market loadings are derived from the first stage time series regression of the Fama and MacBeth (1973) approach using either the unconditional model outlined Equation (12) or the conditional model in Equation (16).

	Past Return					Past Return				
	Low	2	3	4	High	Low	2	3	4	High
Size	Panel A: Avg. Returns (%)					Panel B: β^{RMRF}				
Small	6.2	14.1	16.3	19.3	23.5	0.96	0.77	0.72	0.71	0.88
2	5.6	12.4	15.3	15.5	20.4	1.17	0.93	0.86	0.87	1.07
3	7.3	12.8	12.6	15.2	19.8	1.21	0.94	0.86	0.88	1.09
4	7.4	11.9	12.9	14.1	18.5	1.24	0.98	0.89	0.88	1.06
Large	7.0	10.8	10.0	11.9	14.1	1.28	1.03	0.96	0.95	1.08
	Panel C: β_{Down}^{RMRF}					Panel D: β_{Up}^{RMRF}				
Small	1.04	0.82	0.74	0.71	0.85	0.82	0.70	0.68	0.72	0.93
2	1.28	0.99	0.89	0.88	1.04	0.98	0.82	0.80	0.85	1.11
3	1.35	1.00	0.91	0.89	1.05	1.00	0.83	0.80	0.86	1.15
4	1.38	1.06	0.93	0.88	1.00	1.01	0.86	0.84	0.89	1.16
Large	1.44	1.09	0.98	0.91	0.98	1.02	0.93	0.94	1.02	1.24

Table 8: **Technical Indicators, Volatility Risk and Liquidity Risk.** Panel A reports ordinary least squares estimates for the time series regressions of monthly factor volatilities on the respective average trend-based signals, both during month t , and the lagged monthly factor volatilities (see Equation (18)). Panel B reports ordinary least squares estimates for the same regression using the aggregate liquidity measure of Pastor and Stambaugh (2003) instead of monthly factor volatilities. The Newey and West (1987) robust t -statistics are in parentheses. R^2 statistics are in percentages. The sample period is from April 1964 to December 2015.

	α	β	ψ	R^2
Panel A: Factor Volatility				
RMRF	-1.10 (-10.54)	-0.19 (-6.17)	0.63 (19.43)	52.8
SMB	-1.39 (-9.58)	-0.03 (-1.33)	0.64 (17.24)	42.1
HML	-1.05 (-8.28)	-0.01 (-0.49)	0.74 (24.86)	55.2
RMW	-0.99 (-7.24)	-0.04 (-1.57)	0.77 (25.62)	59.7
CMA	-1.06 (-8.28)	0.01 (0.50)	0.76 (26.66)	57.8
WML	-0.91 (-8.84)	0.12 (4.07)	0.79 (31.67)	62.6
Panel B: Aggregate Liquidity				
RMRF	-0.06 (-9.34)	0.05 (7.24)	0.01 (0.10)	11.8
SMB	-0.03 (-7.23)	0.01 (2.78)	0.11 (2.02)	2.8
HML	-0.02 (-5.53)	-0.01 (-1.26)	0.12 (2.22)	1.8
RMW	-0.03 (-6.48)	0.00 (0.81)	0.12 (2.25)	1.7
CMA	-0.02 (-5.62)	-0.01 (-1.73)	0.12 (2.15)	2.0
WML	-0.02 (-4.19)	-0.01 (-1.33)	0.13 (2.26)	1.8

Table 9: **Technical Indicators and Investor Sentiment.** Panel A reports ordinary least squares estimates for time series Logit regressions of the average trend-based signal on the orthogonalized sentiment-change index of Baker and Wurgler (2006), $\Delta\text{SENT}^\perp$, both during month t . Panel B reports ordinary least squares estimates for the same regression, using the previous month-end orthogonalized sentiment level, $\text{SENT}^\perp$, instead of the sentiment-change index. Newey and West (1987) robust t -statistics are in parentheses. In addition to the regression coefficients, I also report the marginal effects (ME) of the slope coefficients. The sample period is from September 1965 to December 2015.

	α	β	ME
Panel A: Average Trend-based Signals on $\Delta\text{SENT}_t^\perp$			
RMRF	0.48 (5.41)	0.27 (2.90)	0.063
SMB	0.10 (1.20)	0.22 (2.47)	0.054
HML	0.43 (4.85)	-0.27 (-3.00)	-0.065
RMW	0.62 (6.91)	-0.15 (-1.68)	-0.035
CMA	0.52 (5.85)	-0.29 (-3.19)	-0.068
WML	0.90 (9.55)	0.08 (0.80)	0.016
Panel B: Average Trend-based Signals on $\text{SENT}_{t-1}^\perp$			
RMRF	0.59 (6.86)	-0.20 (-2.30)	-0.045
SMB	0.09 (1.04)	-0.48 (-5.40)	-0.121
HML	0.33 (3.94)	0.13 (1.56)	0.032
RMW	0.62 (7.14)	0.42 (4.65)	0.096
CMA	0.45 (5.36)	0.20 (2.31)	0.047
WML	0.92 (10.21)	0.16 (1.71)	0.032

Table A1: **Pairwise Correlations.** This table presents the pairwise correlations between the trend-based signals (Panel A) and the returns on trend-based dynamic factor strategies (Panel B). The sample period is from May 1969 to December 2015.

	SMB	HML	RMW	CMA	WML	SMB	HML	RMW	CMA	WML
	Panel A: Trend-based Signals					Panel B: Trend-based Dynamic Strategies				
RMRF	0.25	-0.12	-0.08	-0.17	0.03	-0.07	-0.14	0.06	-0.11	0.18
SMB		-0.02	-0.23	-0.09	-0.06		0.05	-0.13	0.04	0.11
HML			-0.19	0.41	-0.11			0.06	0.57	0.05
RMW				-0.21	0.00				0.12	0.21
CMA					-0.13					0.16

Table B1: **Long Leg and Short Leg Analysis.** This table examines TI's predictive ability separately for the long leg and short leg of each risk factor. Reported are the slope coefficient (in percentages) and the R^2 statistic of the bivariate predictive regression model given in Equation (4). p -values are based on a one-sided (upper-tail) wild bootstrap. R^2_{OOS} is the Campbell and Thompson (2008) out-of-sample R^2 statistic with p -values based on the Clark and West (2007) *MSFE-adjusted* statistic. ΔCER is the gain in the annualized certainty equivalent return, if an investor uses the trend-based dynamic factor strategy instead of the historical average based strategy to allocate between the respective risk factor and a risk-free asset. ΔSR is the difference between the annual Sharpe ratio of the trend-based dynamic factor strategy and the historical average based strategy. ΔMD is the reduction in the maximum drawdown, defined as the largest negative return over the sample period from a peak to a bottom (in percentages), of the trend-based dynamic factor strategy relative to the historical average based strategy. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from April 1964 to December 2015.

	β (%)	R^2_{IS}	R^2_{OOS}	ΔCER (%)	ΔSR	ΔMD (%)
Panel A: Long Leg						
SMB	18.33*** (3.06)	0.09	0.07***	7.24***	0.55***	21.8
HML	11.67** (1.91)	0.04	0.02**	4.52***	0.33**	15.1
RMW	9.84** (1.65)	0.03	0.00**	3.94***	0.33**	15.7
CMA	11.39** (1.97)	0.04	0.02**	4.22***	0.32**	14.7
WML	14.23** (2.27)	0.05	0.03***	4.77***	0.30**	14.9
Panel B: Short Leg						
SMB	8.60* (1.51)	0.02	0.00*	2.61***	0.29**	9.7
HML	11.45** (1.98)	0.04	0.01**	2.69**	0.41***	19.7
RMW	20.16*** (3.35)	0.10	0.08***	6.00***	0.60***	19.8
CMA	13.80** (2.25)	0.05	0.03***	3.76***	0.50***	23.0
WML	17.41*** (2.73)	0.07	0.04***	3.62***	0.75***	25.6

Table C1: **Robustness Checks.** This table shows the performance of the different technical indicators and parameterizations introduced in Section 2.2. Panel A reports the slope coefficient (in percentages) of the bivariate predictive regression model given in Equation (4). p -values are based on a one-sided (upper-tail) wild bootstrap. Panel B reports the Campbell and Thompson (2008) out-of-sample R^2 statistic, with p -values based on the Clark and West (2007) $MSFE$ -adjusted statistic. Panel C documents the gain in the annualized certainty equivalent return, ΔCER , (in percentages) if an investor uses the trend-based dynamic factor strategy instead of the historical average based strategy to allocate between the respective risk factor and a risk-free asset. Panel D shows the maximum drawdown (MD) (in percentages), defined as the largest negative return over the sample period from a peak to a bottom of the trend-based dynamic factor strategy. Panel E reports the annual turnover which is defined as the average number of times wealth is traded each year. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from May 1969 to December 2015.

	$MA_{1,20}$	$MA_{1,40}$	$MA_{1,120}$	$MA_{1,180}$	$MA_{1,240}$	$MA_{20,180}$	$MA_{20,240}$	MOM_{180}	MOM_{240}	PCA_1	PCA_2
Panel A: β (%)											
RMRF	12.35***	12.37***	10.25**	9.93**	9.88**	1.83	2.41	6.68	5.80	3.86*	8.96**
SMB	20.98***	18.95***	10.87***	9.42***	6.49***	4.81**	5.75***	5.32**	4.94**	4.47***	14.60***
HML	25.93***	23.35***	14.47***	14.03***	14.24***	9.37***	11.07***	9.82***	7.91***	6.54***	16.47***
RMW	15.01***	14.12***	9.19***	7.86***	7.25***	4.52***	5.76***	6.30***	5.81***	4.12***	9.79***
CMA	16.65***	14.87***	8.69***	8.02***	6.12***	4.14***	3.35**	4.76***	4.68***	3.66***	11.67***
WML	40.40***	26.89***	14.20***	12.19***	10.81***	-1.38	4.57*	8.66***	7.83**	6.59***	26.34***
Panel B: R^2_{OOS} (%)											
RMRF	0.01***	0.02***	0.01**	0.01**	0.01*	-0.02	-0.02	-0.01	-0.02	0.01*	-0.01***
SMB	0.36***	0.30***	0.07***	0.04***	-0.02***	-0.04**	-0.02**	-0.02**	-0.04**	0.05***	0.24***
HML	0.83***	0.67***	0.27***	0.25***	0.25***	0.10***	0.15***	0.12***	0.07***	0.34***	0.48***
RMW	0.63***	0.56***	0.23***	0.16***	0.13***	0.03*	0.07***	0.09***	0.08***	0.26***	0.41***
CMA	0.67***	0.56***	0.19***	0.15***	0.08***	0.03**	0.00*	0.04***	0.03**	0.19***	0.48***
WML	0.89***	0.42***	0.11***	0.07***	0.05***	-0.01	0.00	0.03*	0.02*	0.12***	0.62***
Panel C: ΔCER (%)											
RMRF	4.16***	4.50***	3.54***	3.02***	2.46***	-0.10	-0.10	0.93*	0.79	2.18**	3.26**
SMB	7.23***	6.66***	3.72***	3.06***	1.77**	1.27	1.50*	1.11	1.14	2.13**	5.78***
HML	8.66***	7.73***	4.61***	4.66***	4.73***	2.21**	3.18***	2.79***	1.83**	4.95***	7.60***
RMW	5.28***	5.02***	2.93***	2.48***	2.13***	0.44	0.91**	1.52***	1.56***	2.56***	4.58***
CMA	4.93***	4.36***	2.10***	1.75***	0.70	-0.01	-0.03	0.12	-0.09	1.30**	3.95***
WML	10.60***	5.98***	0.58	-0.08	-0.43	0.07	-0.11	-0.32	-0.30	-0.30	9.61***
Panel D: MD (%)											
RMRF	42.46	34.11	19.46	20.81	19.54	23.83	26.10	28.34	26.69	16.14	43.45
SMB	23.69	25.21	33.04	34.07	31.71	30.59	31.20	30.19	31.72	32.78	26.22
HML	26.42	19.26	23.98	22.23	24.39	24.82	24.82	35.57	25.99	21.74	24.12
RMW	16.60	14.31	18.49	16.00	18.60	18.02	16.73	23.21	16.45	20.26	14.13
CMA	11.81	12.52	17.50	19.76	22.06	25.69	25.46	22.12	22.20	21.78	16.98
WML	30.88	40.51	43.28	42.53	41.70	46.65	44.71	41.34	40.84	43.67	36.36
Panel E: Annual Turnover											
RMRF	22.17	14.08	6.59	4.29	3.43	1.27	1.23	2.48	2.05	5.55	19.30
SMB	26.33	16.70	8.69	6.43	6.24	2.11	1.97	5.81	6.13	6.69	23.69
HML	22.07	14.51	6.69	5.70	2.94	2.26	1.41	3.67	2.84	3.88	18.81
RMW	23.29	14.84	6.83	6.27	5.14	2.15	2.05	5.58	4.43	5.23	20.69
CMA	23.29	14.26	8.31	5.52	5.51	1.30	0.74	3.28	3.64	5.32	18.89
WML	19.62	13.31	5.37	3.71	3.06	0.22	0.24	1.32	1.53	4.03	16.61

Table C2: **Combination Forecasts.** This table reports the out-of-sample performance of technical indicators using the five forecast combination methods proposed in Rapach et al. (2010). Specifically, these methods combine the nine individual return forecasts based on the nine technical indicators outlined in Section 2.2 and Equation (6) by either using the average forecast (Mean), the median forecast (Median), the trimmed average, which excludes the lowest and highest forecast (Trimmed Mean), or a method, which determines the combination weights for each individual forecast based on their historical forecasting performance, i.e. the discount mean square prediction errors (DMSPE) over the holdout out-of-sample period discounted with a factor θ . This discount factor θ is either equal to one (no discounting) or below one (greater weight is attached to the recent forecast accuracy). All individual forecasts are estimated recursively using only data available at the end of the forecast formation period t . R^2_{OOS} is the Campbell and Thompson (2008) out-of-sample R^2 statistic. ΔCER is the gain in the annualized certainty equivalent return, if an investor uses the trend-based dynamic factor strategy instead of the historical average based strategy to allocate between the respective risk factor and a risk-free asset. Both measures are in percentages. The sample period is from April 1964 to December 2015.

	Mean		Median		Trimmed Mean		DMSPE $_{\theta=1.0}$		DMSPE $_{\theta=0.9}$	
	R^2_{OOS}	ΔCER	R^2_{OOS}	ΔCER	R^2_{OOS}	ΔCER	R^2_{OOS}	ΔCER	R^2_{OOS}	ΔCER
RMRF	0.05**	2.87***	0.03**	2.37***	0.05**	2.96***	0.04**	2.51***	0.05**	2.57***
SMB	0.31***	4.30***	0.07**	2.18**	0.25***	3.60***	0.27***	3.88***	0.31***	4.18***
HML	0.49***	5.45***	0.34***	4.91***	0.41**	5.07***	0.48***	5.29***	0.52***	5.60***
RMW	0.35***	3.76***	0.19***	2.59***	0.31***	3.50***	0.35***	3.65***	0.36***	3.78***
CMA	0.36***	2.12***	0.18***	1.50***	0.29***	1.95***	0.35***	1.90***	0.37***	2.01***
WML	0.32***	2.06***	0.14***	0.91*	0.23***	1.15**	0.32***	2.15***	0.36***	2.73***

Table C3: **Alternative Risk Adjustment Coefficients.** This table compares the performance of the trend-based dynamic factor strategy (TRS) to the historical average based strategy (HA) for different risk adjustment coefficients, γ , used in Equation (8). The respective out-of-sample forecasts are estimated recursively, using only data available at the end of the portfolio formation period t . CER is the annualized certainty equivalent return for the investor (in percentages). The Sharpe ratio is annualized and calculated as the mean portfolio return in excess of the risk-free rate divided by its standard deviation. MD is the maximum drawdown defined as the largest negative return over the sample period from a peak to a bottom (in percentages). Turnover is the average number of times wealth is traded each year. Δ denotes the difference in the respective performance measure between the trend-based dynamic factor strategy and either the historical average based factor strategy or the static factor strategy. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from May 1969 to December 2015.

	CER (%)			Sharpe Ratio			MD (%)		Turnover	
	TRS _j		HA _j	TRS _j		HA _j	TRS _j	HA _j	TRS _j	HA _j
	CER	CER	Δ CER	SR	SR	Δ SR				
Panel A: $\gamma = 0.5$										
RMRF	13.18	8.49	4.69	0.58	0.26	0.32	45.1	69.5	6.69	1.08
SMB	8.22	5.56	2.66	0.39	0.11	0.28	34.3	49.6	6.67	0.16
HML	14.17	11.15	3.03	0.95	0.53	0.42	22.2	58.0	5.62	0.00
RMW	11.36	9.88	1.48	0.93	0.60	0.33	23.1	48.8	6.41	0.97
CMA	11.87	11.01	0.86	0.89	0.68	0.20	19.8	26.0	5.53	0.25
WML	16.42	16.59	-0.18	0.77	0.70	0.06	56.2	79.1	3.59	0.00
Panel B: $\gamma = 1.5$										
RMRF	9.93	5.76	4.17***	0.55	0.23	0.32**	35.2	40.7	5.59	0.83
SMB	7.39	3.67	3.72***	0.39	0.01	0.38***	34.3	49.1	6.64	0.49
HML	13.23	8.89	4.34***	0.95	0.49	0.46***	22.2	57.7	5.71	0.10
RMW	10.87	8.57	2.31***	0.93	0.59	0.33***	23.1	48.7	6.44	0.88
CMA	11.23	9.84	1.38**	0.88	0.66	0.22**	19.8	26.0	5.52	0.31
WML	14.60	14.50	0.10	0.84	0.80	0.04	49.9	64.7	3.97	0.07
Panel C: $\gamma = 5$										
RMRF	6.51	5.05	1.46**	0.54	0.20	0.34***	14.8	16.5	2.37	0.33
SMB	5.48	3.31	2.17***	0.41	-0.06	0.47***	21.2	40.9	4.84	0.43
HML	9.81	5.96	3.85***	0.97	0.52	0.45***	20.2	30.6	5.18	0.37
RMW	8.70	6.80	1.91***	0.94	0.61	0.33***	14.5	31.6	5.90	0.56
CMA	8.79	6.96	1.84***	0.89	0.61	0.28**	18.8	21.5	5.27	0.29
WML	11.15	10.91	0.24	1.07	1.05	0.02	37.9	28.8	3.33	0.25

Table D1: **Tests of Exclusion Restrictions for the Trend-based Conditional Asset Pricing Models.** This table reports heteroskedasticity-consistent χ^2 -statistics for tests of parameter restrictions for the trend-based CAPM, the trend-based Fama-French three-factor model and the trend-based Fama-French five-factor model. p -values are based on a one-sided (upper-tail) wild bootstrap. The trend-based conditional models are specified as outlined in Equation (16):

$$r_{i,t+1} - r_{f,t+1} = \alpha_i + \beta_{i,t}^j r_{j,t+1} + \dots + \epsilon_{i,t+1},$$

where $r_{i,t+1}$ is the return on test asset i during period $t+1$, $r_{f,t+1}$ is the rate on the risk-free asset and $r_{j,t+1}$ is the return on risk factor j , and $TI_{j,t}$ is the trend-based signal on risk factor j . The χ^2 -statistics correspond to the exclusion restrictions of $\beta_{i,Up-Diff}^j = 0$ for all considered risk factors j . As test assets, I use either the 25 size and momentum sorted portfolios (Panel A) or the 30 industry-sorted portfolios (Panel B). *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is from April 1964 to December 2015.

Panel A: 25 Size and Mom. Sorted Portfolio				Panel B: 30 Industry-sorted Portfolios			
Portfolio	CAPM	FF3	FF5	Portfolio	CAPM	FF3	FF5
11	57.70***	432.13***	479.29***	Food	7.16***	11.52***	6.47***
12	27.89***	235.07***	325.39***	Beer	21.78***	24.23***	14.05***
13	9.03***	77.14***	142.53***	Smoke	42.30***	48.93***	37.69***
14	0.51***	19.07***	73.33***	Games	16.61***	21.02***	28.45***
15	12.65***	57.34***	64.40***	Books	2.03**	4.01	14.76***
21	76.84***	538.02***	677.98***	Hshold	4.66***	6.19***	3.23
22	56.24***	383.50***	447.04***	Clths	3.87**	3.43**	18.01***
23	22.31***	130.05***	185.92***	Hlth	33.05***	31.83***	42.94***
24	2.46***	27.68***	59.25***	Chems	3.62**	12.32***	51.26***
25	14.00***	87.45***	93.64***	Txtls	13.37***	17.77***	38.98***
31	120.73***	446.28***	567.04***	Cnstr	4.34**	8.48***	30.72***
32	67.20***	337.96***	367.98***	Steel	53.00***	53.50***	54.68***
33	43.66***	207.89***	272.41***	FabPr	16.48***	34.96***	41.46***
34	4.28***	10.10***	40.11***	ElcEq	0.00	18.74***	41.40***
35	34.63***	182.19***	204.08***	Autos	1.49	1.30	8.10*
41	116.23***	336.06***	457.23***	Carry	0.00**	3.49**	21.68***
42	120.82***	386.94***	421.90***	Mines	4.06**	4.79***	24.77***
43	50.46***	112.73***	150.73***	Coal	7.30***	29.91***	33.74***
44	0.21	9.24***	29.16***	Oil	0.41	7.04***	21.12***
45	96.19***	261.09***	362.23***	Util	6.28***	20.67***	31.52***

Table D1: Tests of Exclusion Restrictions for the Trend-based Conditional Asset Pricing Models. Continued.

Panel A: 25 Size and Mom. sorted Portfolio				Panel B: 30 Industry-sorted Portfolios			
Portfolio	CAPM	FF3	FF5	Portfolio	CAPM	FF3	FF5
51	116.11***	255.48***	384.26***	Telcm	6.52***	19.42***	23.76***
52	91.32***	260.84***	282.72***	Servs	0.05	4.98**	12.35***
53	7.75***	60.28***	94.05***	BusEq	0.00	37.22***	45.89***
54	55.41***	111.61***	123.73***	Paper	0.01	11.06***	28.87***
55	288.37***	455.37***	512.62***	Trans	1.05	7.26***	16.15***
				Whlsl	2.74***	9.12***	22.17***
				Rtail	2.24**	3.33	5.08
				Meals	0.72**	9.17***	24.80***
				Fin	29.89***	81.73***	71.07***
				Other	24.51***	33.36***	40.50***

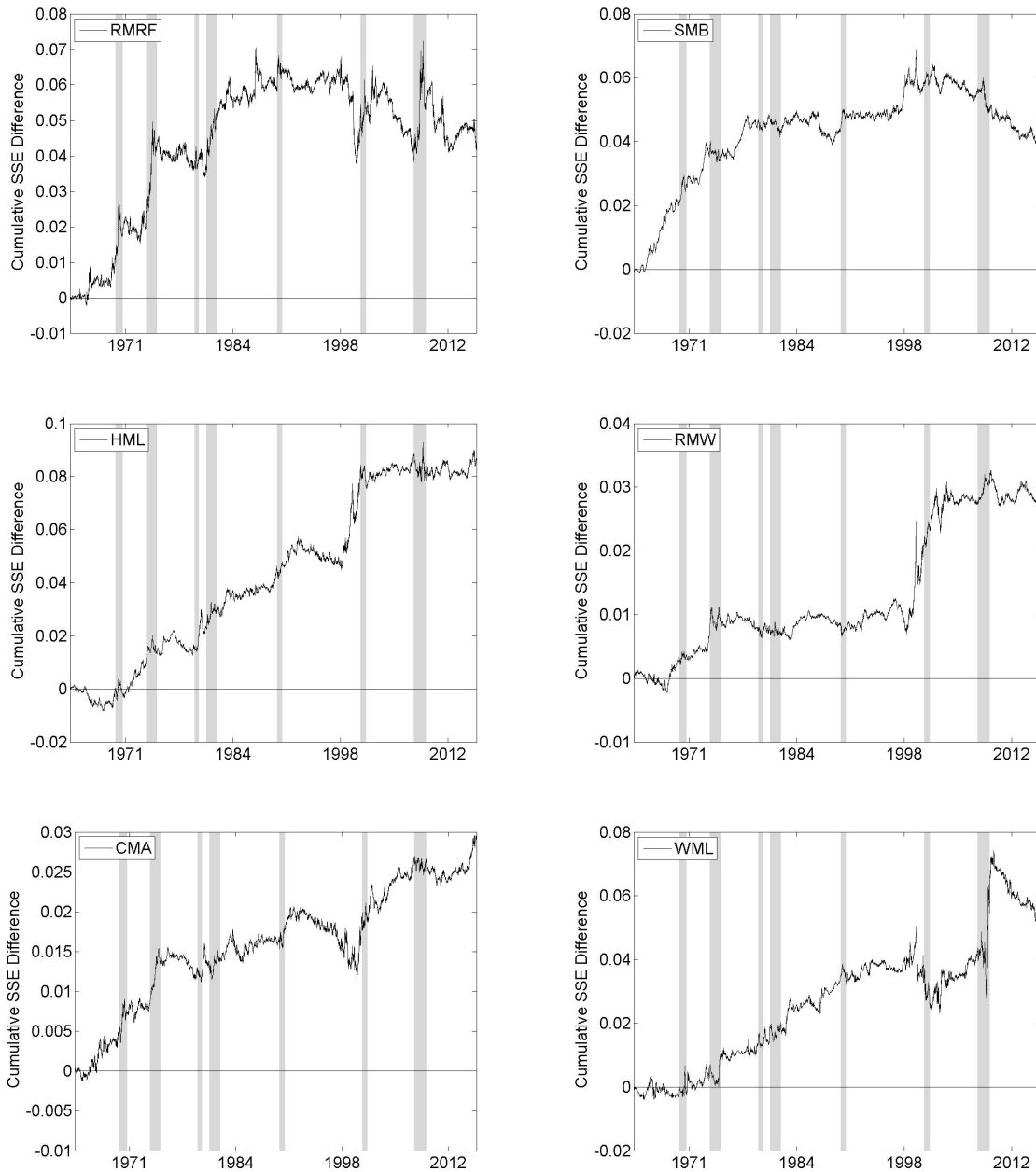


Fig. 1. **Predictability Over Time.** This figure depicts the difference between the cumulative squared forecast errors (CSFE) for the historical average benchmark and trend-based forecasts. An increase in the line indicates a better performance of the trend-based forecast; a decrease indicates a better performance of the historical average. Both forecasts are estimated recursively based on information available at the end of the forecast formation period t . Vertical bars correspond to NBER-dated recessions.

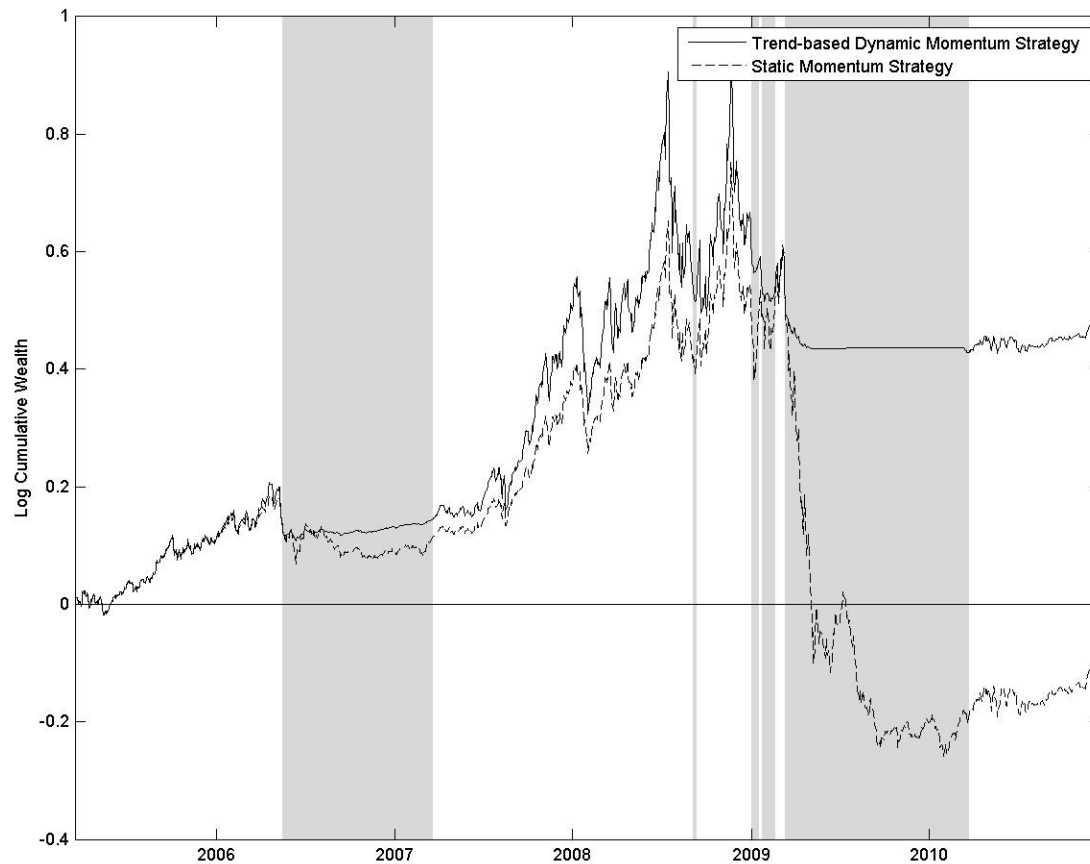


Fig. 2. **Performance during the "Momentum Crash" of 2009.** This figure delineates the log cumulative wealth of the trend-based dynamic momentum strategy (solid line) and the static momentum strategy (dashed line) for an investor who allocates between WML and a risk-free assets and uses a risk adjustment coefficient, γ , equal to 2.5. I assume that the investor begins with \$1 and reinvests all proceeds. Vertical bars depict downtrends in WML indicated by the trend-based indicator.